

SOFTWARE REQUIREMENTS AND DESIGN DOCUMENT AND LAUNCH PROPOSAL

Lintel

Property management and real estate platform by Codevertex: estate management, unit sales, rental and portfolio management, and Keybazaar, its public marketplace. Launching with Shaba Village, Syokimau.

LAUNCH TENANT

Shaba Village
Estate management

PREPARED BY

Codevertex Africa Limited
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DATE

6 October 2026

RELEASE 1

14 working weeks
Launch February 2027

LAUNCH INVESTMENT

KES 250,000 build
From KES 10,000 per month

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Document control

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Role	Party	Responsibility
Product owner, platform	Codevertex	Product roadmap, multi-tenant design, pricing, marketplace policy
Product owner, launch tenant	Shaba Village estate management	Shaba scope approval, business rules, by-law and tariff decisions, data supply, legal sign-off
Finance lead, launch tenant	Shaba Village	Charge rates, opening balances, bank and paybill set-up, approval limits
Delivery and technical leads	Codevertex	Planning, architecture, code quality, security, releases

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Executive summary

Lintel, by Codevertex, is a multi-tenant software-as-a-service platform for the Kenyan real estate market, built on the same shared services that already run Codevertex's commerce, finance and HR products. One platform serves three kinds of business, each as an isolated tenant with its own users, properties, money and brand:

- **Estate operators and developers** such as Shaba Village, who sell homes and apartments outright or by instalments and then manage the estate and bill owners for water, service charge and shared services.
- **Property management firms** that let and manage buildings for many landlords: they list units, onboard tenants on signed leases, collect rent and utilities, arrange repainting and repairs, assign caretakers and managers to each property, and remit to landlords with statements.
- **Landlords, developers and agents** who manage their own portfolios or advertise property for rent or sale.

A public marketplace, Keybazaar, added in Release 3, lets house hunters and businesses browse homes, offices, shops, warehouses and land for rent or sale. It draws on the vacancies of properties already managed on the platform, so listings are verified at the source and a successful enquiry flows straight into a lease or a sale contract.

The minimum viable product, Release 1, is deliberately tight. It builds the multi-tenant core and the full estate management workflow, and launches Shaba Village as the first tenant within 14 weeks. Rental and portfolio management for property managers follows in Release 2, and the marketplace in Release 3, on the same data model, so nothing built for Shaba is rebuilt later.

What Shaba Village gets at launch

- One balance per unit and fund, visible to each owner on their phone, with statements on demand.
- Paybill payments matched to the right unit and receipted within a minute.
- Monthly bills, including metered water with photo evidence, issued for every unit in one run.
- Instalment schedules, reminders and purchase statements from reservation to handover and title.
- Sale agreements, handover forms, clearance certificates and notices generated from templates and stored against each unit.
- Service provider contracts, schedules, guard posts and patrols, with vendor invoices adjusted for missed service.
- Visitor passes and gate logs that collect only what the law allows.
- Estate staff and casual workers paid through the ERP payroll; sales and estate money kept in separate ledgers.

Engagement summary

14 weeks Release 1 and Shaba launch, February 2027	KES 250,000 Fixed launch build price, inclusive of taxes	KES 20,000 Recommended monthly tier: subscription, hosting, support	3 releases Estate management, rental management, marketplace
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SECTION 1

Introduction

1.1 Purpose

This document specifies the requirements and design of Lintel across its releases, the delivery plan for Release 1, and the commercial terms for Shaba Village as launch tenant. It is the basis for development, testing and acceptance, and replaces the earlier property management requirements document.

1.2 Product vision

Kenyan property businesses run on spreadsheets, paybill statements, WhatsApp groups and paper files. Rent and service charge are reconciled by hand, deposits disappear into general accounts, repairs are chased by phone, and house hunters lose money to fake listings. Lintel gives each business one system of record for its properties, people, agreements and money, and gives the public one place, Keybazaar, to find property that is real, available and verified.

1.3 The name

The platform is called **Lintel**. A lintel is the load-bearing beam above every doorway: unseen by most people, it carries the weight so the door can open. That is the job of the platform for a property business, holding up the money, agreements, people and services behind every unit. The public marketplace in Release 3 is **Keybazaar**: a bazaar where keys change hands, whether a family is renting its next home, a business is taking an office, or an investor is buying land. A seeker who finds a home on Keybazaar then pays rent or service charge through the Lintel portal of the business that let it. Following other Codevertex products such as Books, whose service is treasury-api, both brands sit on functional service names: property-api, property-ui and marketplace-ui.

1.4 Launch tenant: Shaba Village

Shaba Village is a residential estate in Syokimau, off Mombasa Road, in the Mavoko area of Machakos County. It comprises homes and apartments, including two and three bedroom units, sold outright or on deposit and instalments. Estate management supplies water from the Mavoko Water and Sewerage Company connection and two boreholes through an underground reservoir and unit tanks, and runs perimeter security, 24-hour guarding, landscaping, cleaning, garbage collection, sewer and parking, all billed to owners every month. Shaba's needs define Release 1; its configuration is described in [Estate management: Shaba Village configuration](#).

1.5 Scope by release

R1	MVP and Shaba Village launch Multi-tenant core, properties and units, staff per property, estate billing, utilities, sales and instalments, owner portal, providers, works, gate, ERP staff, reports	14 weeks
R2	Rental and portfolio management Landlord clients and mandates, tenant onboarding, leases, rent and deposits, inspections, turnover repairs, landlord statements and remittances, commercial leases	Q2 2027
R3	Public marketplace Listings for rent and sale (homes, offices, shops, land), search and map, enquiries, viewings, applications, verified listers, featured listings	Q3 2027
R4	Extensions Native apps, smart meters, e-signatures, tenant credit checks, owner voting, AI assistant	Later

Figure 1. Release plan

Area	Release 1 (MVP)	Later releases
Platform	Multi-tenant core, self-service tenant settings, plans and limits, per-tenant branding and domain, platform administration	Self-service sign-up with trial (R2); marketplace operations (R3)
Properties	Properties, blocks, units, staff assignment per property, owners, occupants, documents	Landlord client portfolios and management mandates (R2); listings (R3)
Sales	Price lists, reservations, sale agreements, instalments, handover, title tracking	Off-plan milestone certificates and buyer marketplace leads (R3)
Billing	Service charge, utilities, levies, meters, billing runs, M-Pesa matching, arrears, statements	Rent, deposits, escalations, landlord remittances (R2)
Leasing	Tenants as occupants of owner units, with bill-to assignment	Full lease lifecycle: applications, leases, inspections, renewals, move-out, deposit refunds (R2)
Operations	Service providers, SLAs, guard posts, patrols, work orders, preventive maintenance, gate and visitors	Turnover repaint and repair projects (R2); smart meters (R4)
Documents	Template library, generated agreements and notices, uploads, signatures by acceptance or upload	Lease and mandate templates (R2); advanced electronic signatures (R4)
Marketplace	Not in Release 1	Public search, map, enquiries, viewings, applications, verified listers, featured listings (R3)

Excluded from all releases: conveyancing and legal advice, mortgage origination, holding of client money by Codevertex, and short-stay holiday booking.

1.6 Definitions

Term	Definition
Tenant (platform)	A business subscribed to the platform, with its own isolated data, users and money. To avoid confusion, the document calls people who rent property lessees or renters wherever the two meanings could meet
Portfolio	A group of properties: a property manager keeps one per landlord client
Property	An estate, building, plot or complex, made up of units
Unit	A house, apartment, office, shop, bay or plot with its own code, for example B07
Owner, buyer	The registered or contracted purchaser of a unit
Landlord	A property owner who lets units, directly or through a property manager
Lessee, occupant	A person or business renting a unit; any person living in or using it
Management mandate	The agreement under which a property manager acts for a landlord
Unit entitlement	A unit's share of common costs
Service charge, sinking fund	Monthly contribution to running costs; reserve for major repairs
Billing run	The monthly process that issues invoices for all units of a property
Listing	A public advertisement of a unit or land for rent or sale
Management corporation	The owners' body corporate under the Sectional Properties Act, 2020
SLA, work order	Measurable service target in a contract; a recorded repair or maintenance job
Paybill, STK push	M-Pesa business number; payment prompt sent to the payer's phone
ODPC, DPIA, EARB	Data Protection Commissioner; impact assessment; Estate Agents Registration Board
eTIMS, WHT, MRI	KRA tax invoice system; withholding tax; monthly residential rental income tax

1.7 References

- Property Management and Real Estate Services Platform requirements document, version 1.0, October 2026 (superseded)
- Codevertex platform documentation: microservice architecture and tenant isolation tiers, cross-service data ownership, treasury invoicing, payments, escrow and approvals, M-Pesa integration, ERP payroll, notifications, subscription gating
- Sectional Properties Act, 2020; Land Registration Act, 2012; Stamp Duty Act; Law of Contract Act as amended by the Business Laws (Amendment) Act, 2020; Estate Agents Act
- Data Protection Act, 2019 and Regulations, 2021; ODPC draft Guidance Note for the Private Security Sector, 2025
- KRA guidance on monthly rental income tax and withholding on rent; Mavoko Water and Sewerage Company tariff, 2023/24 to 2025/26
- Web Content Accessibility Guidelines 2.2; OWASP Application Security Verification Standard 4

SECTION 2

Market and tenant types

Who the platform serves, what each segment needs, and how comparable products have shaped the design.

2.1 Tenant types

Tenant type	Examples	Core needs	Release
Estate operator or developer	Gated estates, apartment developments, sectional property schemes such as Shaba Village	Unit sales and instalments, owner billing, metered water, service providers, gate security, AGM transparency, handover to the management corporation	R1
Property management firm	Letting and management agencies managing residential and commercial buildings for many landlords	Landlord portfolios and mandates, staff per property, tenant onboarding, leases, rent and utilities, deposits, repairs and repainting, landlord statements and remittances, management fees	R2
Self-managing landlord	Owners of one or more apartment blocks, commercial buildings or rental plots	As above without the client layer: rent collection, repairs, caretakers, tax records	R2
Agent or developer advertising	Registered estate agents, developers selling off-plan, land sellers	Verified listings, enquiries, viewings, lead handling	R3
Service provider	Security, cleaning, garbage, maintenance contractors	Schedules, work orders, evidence, invoicing and payment across the tenants they serve	R1
Public	House hunters, buyers, businesses seeking offices or shops, diaspora investors	Real, current listings with prices, photos, location and a safe way to enquire and apply	R3

2.2 Comparable systems and practice

Product or practice	What it shows	How it shaped the design
Kenyan rent tools (paybill-first rent collection)	Automated paybill matching per unit and SMS or WhatsApp reminders are what reduce arrears locally	Unit-coded paybill accounts, automatic matching, escalating reminders
AppFolio, Buildium (US)	Association and rental management with vendor portals, work orders, violations and owner statements in one product	Vendor portal, owner and landlord statements, one work order model for estates and rentals
Yardi, MRI (global)	Commercial leases need escalations, service charge recovery and tenant statements	Commercial lease terms in R2: escalation, VAT on commercial rent, service charge recovery
Gated estate access apps	QR passes, guard verification and host approval are now expected by residents	Gate tablet with passes, walk-in approval and offline mode

Property portals (local and global)	Fake listings and deposit scams are the biggest risk in Kenyan house hunting	Verified listers, listings sourced from managed vacancies, masked contacts, no deposits outside the platform
Kenyan estate disputes	Unaccounted service charge and mixed funds drive litigation by owners	Separate fund ledgers, budget against actual, AGM statements

2.3 Positioning

Local tools collect rent but do not run an estate, sell units or manage providers; global tools are priced in dollars and do not speak M-Pesa, eTIMS or Kenyan tax. Lintel combines both, and it is the only product in the segment that sits on a full accounting, payroll and messaging stack already in production.

SECTION 3

Overall description

3.1 Product perspective

Each customer is a tenant of the Codevertex platform. Three new components, property-ui, property-api and, in Release 3, marketplace-ui, provide the property product. Sign-in, invoicing, M-Pesa, the ledger, approvals, payroll, messaging, plans, leads and maps come from existing platform services, described in [Solution architecture](#). The property service holds references to their records and never keeps a second copy.

3.2 User classes

Role	Description	Main permissions	Sign-in
PLATFORM			
Platform administrator	Codevertex staff	Tenants, plans, marketplace moderation and verification; no access to tenant financial data except under a logged support grant	SSO with MFA
TENANT STAFF			
Tenant administrator	Directors and delegates	All tenant settings, staff, roles, approvals above limit	SSO with MFA
Property manager	Runs one or more assigned properties	Operations, leases, work orders, vendors, notices and reports for assigned properties only	SSO with MFA
Finance officer	Billing, collections and payouts	Charges, billing runs, allocations, adjustments within limits, vendor bills, remittances	SSO with MFA; OTP per approval
Sales or letting officer	Unit sales or lettings	Price lists, reservations, agreements, applications, listings	SSO with MFA
Caretaker	On-site staff at an assigned property	Meter readings, work orders, inspections, service checks, notices	SSO, phone
CUSTOMERS OF THE TENANT			
Owner, buyer	Purchaser of a unit, including diaspora owners	Statements, payments, purchase progress, readings, requests, passes, household, documents	Phone and OTP
Landlord client	Property owner using a property manager (R2)	Portfolio dashboard, statements, remittances, approval of repairs above limit, documents	Phone and OTP
Lessee or occupant	Renter, household member or tenant of an owner	Lease, bills and payments assigned to them, requests, passes, notices	Phone and OTP
FIELD AND PARTNERS			
Guard	Agency guard on duty	Gate tablet only: passes, walk-ins, deliveries, patrols, incidents	Gate PIN on a registered tablet
Vendor supervisor	Agency or contractor representative	Own contracts, schedules, rosters, work orders, invoices, documents	Phone and OTP
PUBLIC (RELEASE 3)			

Visitor	Anyone browsing	Search and view listings	None
Registered seeker	House hunter, buyer or business	Saved searches, enquiries, viewing requests, applications	Phone and OTP
Lister	Verified agent, developer or private owner without a full subscription	Own listings, enquiries, viewings	Phone and OTP, verified

One person can hold different roles in different tenants, for example an owner at Shaba Village who also rents an office managed by another tenant. They sign in once; each tenant sees only its own relationship with them.

3.3 Operating environment

- All users: any modern mobile or desktop browser; installable as a progressive web application
- Gate: an Android tablet at each gate in kiosk mode, with offline operation during network loss
- Hosting: Codevertex Kubernetes platform with PostgreSQL, Redis, NATS JetStream and object storage, behind Cloudflare
- Payments: Safaricom M-Pesa through each tenant's own paybills (Daraja API) and bank transfers recorded against unit accounts
- Messaging: Africa's Talking (SMS), Meta WhatsApp Cloud API, Codevertex mail server (email)

3.4 Constraints

- Money is collected only into each tenant's own paybills and bank accounts. Codevertex never receives tenant or customer funds.
- The property service does not move money. Invoices, payments and payouts are executed by the treasury service.
- Utility supply is never cut off automatically, and nobody is locked out, for arrears. Kenyan courts and tribunals have held self-help disconnection unlawful.
- Electricity is not resold; units use Kenya Power prepaid meters.
- Conveyancing, lease registration and legal notices are performed by advocates; the platform generates drafts, stores signed documents and records progress.

3.5 Assumptions and dependencies for the Shaba launch

- Shaba Village supplies the unit register, owner list, sale agreements, templates of its existing forms and opening balances by week 4, in the import template provided.
- Shaba Village holds, or opens, separate paybills and bank accounts for sale collections and for the estate fund, and grants Daraja API access by week 3.
- Shaba Village confirms the charge schedule, water tariff, penalty rules and allocation order from its by-laws and sale agreements by week 5.
- Service provider contracts and licences are available for loading by week 8.
- Shaba Village appoints a product owner able to decide within 2 working days and supply documents within 3.
- Shaba Village registers with the ODPC as a data controller and engages an advocate to approve the legal pack and document templates.
- Shaba Village provides one Android tablet per gate, with a data connection.
- Messaging, payment provider charges and the domain name are client-borne costs, as set out in [Commercial terms](#).

SECTION 4

Multi-tenancy and SaaS design

Every business on the platform is a tenant. Tenants share the software and infrastructure, never each other's data or money.

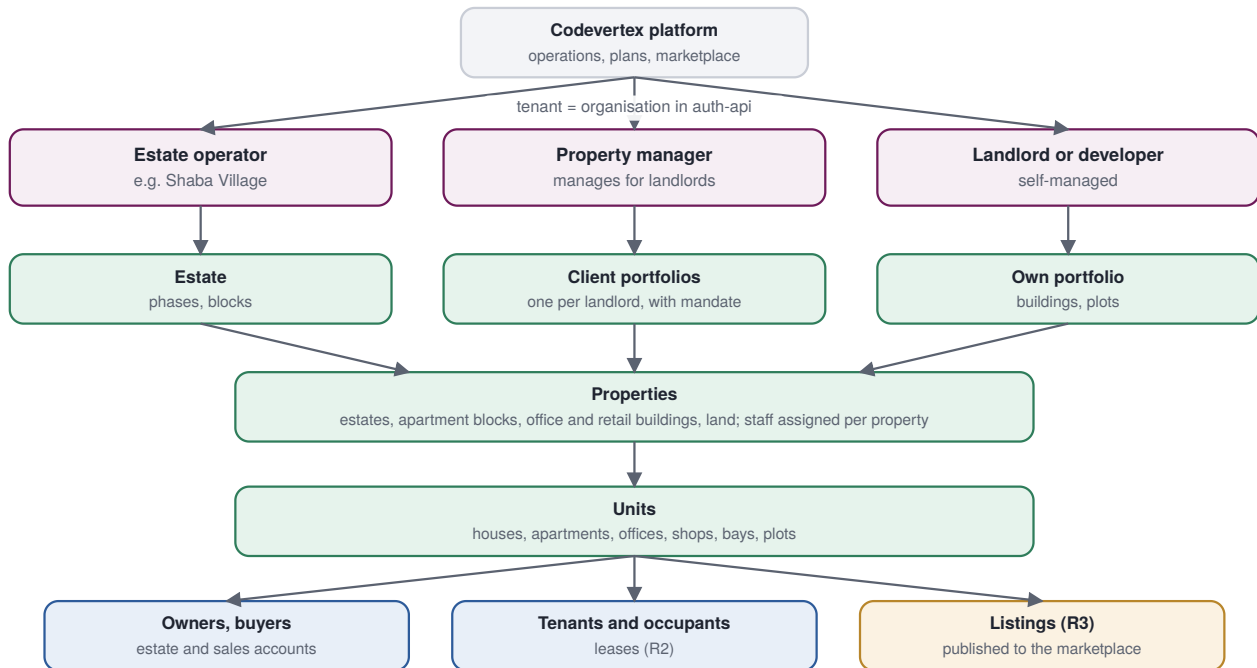


Figure 2. Tenant and property hierarchy

4.1 Tenant model

Level	Owned by	Description
Tenant	auth-api	The organisation: legal name, KRA PIN, branding, contacts, staff, roles and plan. The same tenant record is used by treasury, ERP, notifications and subscriptions, so a property tenant gets accounting, payroll and messaging with no second sign-up.
Portfolio	property-api	A group of properties. An estate operator has one; a property manager has one per landlord client, carrying the management mandate, fee rules and remittance account.
Property	property-api, registered as a branch in auth-api	An estate, building, plot or complex. Each property is also an auth-api branch, so the platform's existing branch-scoped roles assign staff to specific properties.
Block, unit	property-api	Physical structure and the lettable or saleable spaces, each with use (residential, office, retail, industrial, parking, land), size and status.
Parties	property-api, with auth-api user IDs	Owners, buyers, landlords, lessees and occupants linked to units by dated relationships.

4.2 Isolation

Control	Implementation
Data isolation	Shared database with tenant_id on every row and PostgreSQL row-level security, so a missing filter in code still cannot read another tenant's rows. Separate schema or separate database available for large or regulated tenants, the platform's standard isolation tiers.
Request scoping	Tenant taken from the signed token, never from a request parameter alone; branch scope applied for property-level roles.
Money	Each tenant's own paybills, bank accounts and ledgers in treasury. A payment to one tenant's paybill can only ever be matched within that tenant.
Files	Object storage paths prefixed by tenant; signed, short-lived links checked against tenant and role.
Messaging	Tenant sender IDs and WhatsApp numbers; messages carry the tenant's name and contacts.
Fair use	Rate limits and job queues per tenant, so one tenant's billing run or import never slows another.
Cross-tenant identity	A person has one sign-in; each tenant sees only its own relationship. Marketplace profiles are separate from tenant records and shared with a lister only when the seeker enquires.
Exit	Per-tenant export and deletion without touching other tenants.

4.3 Modules and use cases

The platform is a set of modules. Each tenant chooses the modules and use cases it needs, and can set them per property, because one tenant may run a gated estate and a commercial building side by side. The product then behaves as if only those modules exist.

Module	What it adds	Depends on	Rel.
Properties and units	Register, parties, staff per property, documents	Always on	1
Billing and collections	Charge catalogue, billing runs, paybill matching, arrears, statements	Properties	1
Utilities and metering	Meters, reading rounds, tariffs, water balance	Billing	1
Unit sales and instalments	Price lists, reservations, sale contracts, instalments, handover, title	Billing	1
Estate and owners	Owner accounts, sinking fund, levies, AGM pack, clearance certificates	Billing	1
Maintenance	Requests, work orders, preventive schedules, recharges	Properties	1
Service providers	Vendors, contracts, SLAs, visits, vendor portal, invoice checks	Maintenance	1
Gate and access	Passes, gate tablet, guard posts, patrols, incidents	Properties	1
Staff and payroll	Link to ERP staff, payroll and casual payments	ERP plan	1
Communication	Notices, alerts, document library, polls	Properties	1
Residential leasing	Applications, leases, deposits, inspections, renewals, turnover	Billing	2
Commercial leasing	Commercial terms, VAT on rent, service charge recovery, escalations	Residential leasing	2

Landlord portfolios	Mandates, landlord portal, remittances, management fees	Leasing	2
Marketplace publishing	Listings, enquiries, viewings, online applications	Properties; leasing or sales	3
Amenity booking	Halls, pools, courts and visitor parking booking with fees	Billing	4

Use case presets switch on a sensible set of modules in one step, and can then be adjusted:

Use case preset	Modules switched on
Estate operator and developer (Shaba Village)	Properties, billing, utilities, sales, estate and owners, maintenance, providers, gate, staff, communication
Developer, sales only	Properties, billing, sales, communication; marketplace from R3
Owners' association or management corporation	Properties, billing, utilities, estate and owners, maintenance, providers, gate, communication
Residential property manager	Properties, billing, utilities, residential leasing, landlord portfolios, maintenance, providers, communication
Commercial property manager	As residential, plus commercial leasing; gate optional
Self-managing landlord	Properties, billing, utilities, residential leasing, maintenance, communication
Agent, listings only	Marketplace publishing (R3)

4.4 How configuration changes the product

A feature is available only where three conditions meet: the tenant's plan includes it, the tenant has switched the module on, and the property's use case includes it. Releases not yet live are hidden for everyone. The result is applied in every layer:

Layer	Behaviour
Navigation and dashboards	Menus, tiles, quick actions and dashboard widgets are built from the enabled modules; a property manager with no gate module never sees passes or patrols
Forms and records	Fields, tabs and steps of disabled modules are not shown; a unit form shows sale fields only when sales is on, and lease fields only when leasing is on
Onboarding and set-up	The set-up wizard asks only for what the enabled modules need, for example meters only when utilities is on
Customer portals	Owners, lessees and landlords see only the screens that apply to their relationship and the property's modules
Roles and permissions	Permissions of disabled modules cannot be granted; role templates follow the use case
API	Routes of disabled modules return a clear "module not enabled" response; checks run on the server, so hiding a button is never the only control
Background jobs and events	Billing runs, reminders, SLA timers and event consumers run only for tenants and properties with the module on, so nothing is billed, sent or written for unused features
Notifications, reports and exports	Message templates, reports and export columns follow the enabled modules

Switching off	Turning a module off hides it and stops its jobs, but keeps its data read-only and exportable; turning it back on resumes from the same data
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Module and use case settings are cached with the tenant profile and invalidated on change, so the check adds no noticeable latency.

4.5 Configuration over customisation

Everything that differs between businesses is configuration, not code. Shaba Village is configured, not customised; the next estate, property manager or landlord starts from the same product and the same seeded defaults, and changes only what is different about it. Settings are layered, and the most specific level wins:

Level	Set by	Example
Platform default	Codevertex seed data	Standard charge catalogue, unit types, templates, work order categories
Tenant	Tenant administrator	Shaba adds a borehole water charge and renames service charge to estate levy
Property	Property manager or finance	One building charges for lift maintenance; another does not
Unit type or unit	Finance	Three bedroom rate differs from two bedroom; a penthouse has a custom rate

4.6 Charge catalogue

Properties do not all charge for the same services, so charges are a catalogue, not a fixed list. The platform seeds the common charges below. Each tenant switches on the ones it uses, edits their names and rates, and adds its own custom charges at any time, without a release.

Group	Seeded charges
Occupancy	Rent, deposit, reservation fee, move-in fee, key and access card deposit
Estate and building services	Service charge, security levy, cleaning, garbage collection, landscaping, common area power, lift maintenance, generator or backup power, CCTV and access control, pest control
Utilities	Water (metered or flat), borehole water, sewer, water bowser supply, internet, gas
Reserves and levies	Sinking fund, special levy, insurance recovery, land rent and rates recovery
Amenities	Parking bay, visitor parking, swimming pool, gym, clubhouse or hall booking, storage
Recoveries and fees	Repair recharge, cleaning on exit, replacement keys or cards, legal and demand letter fees, late payment charge, bounced cheque fee
Sales	Unit price, deposit, instalment, variation or upgrade, transfer and legal fees recovery

Every charge, seeded or custom, carries the same settings:

Setting	Options
Basis	Fixed amount, per unit type, per square metre, by unit entitlement, metered with block or flat tariff, percentage of another charge, one-off

Frequency	Monthly, quarterly, half-yearly, annual, on event (move-in, move-out, booking), one-off
Applies to	Tenant, selected properties, unit types, units, or parties who opt in (for example a second parking bay)
Bill to	Owner, lessee or occupant, landlord; whether an owner may reassign it
Fund and accounts	Fund (sales, estate, deposits, rent collected for landlords), income or liability ledger account, cost centre
Tax	VAT rate or exempt, eTIMS item code, withholding treatment
Proration	By days, by full month, or none, on start and end dates
Penalty	Whether a late payment charge may apply, grace days, rate, cap
Allocation priority	Order in which payments settle it
Effective dates	Rates versioned by date; issued invoices never change

4.7 Other catalogues

Catalogue	Seeded defaults, extendable per tenant
Property types and unit uses	Estate, apartment block, office, retail, mixed use, industrial, land; house, apartment, office, shop, warehouse, bay, plot
Amenities and features	Borehole, backup generator, lift, pool, gym, parking, CCTV, electric fence, solar water heating, fibre
Document templates	Per tenant type, listed in Documents and agreements
Work order categories and SLAs	Plumbing, electrical, structural, water supply, security systems, common areas, appliances; priority targets
Vendor categories and required documents	Security (PSRA licence), garbage (NEMA licence), cleaning, pest control (PCPB licence), electrical (EPRA licence), contractors (NCA)
Inspection checklists	Rooms and items for move-in, move-out, handover and routine inspections
Pass types, incident and notice categories	As described in the security and communication chapters
Roles and permissions	Tenant administrator, property manager, finance, sales or letting, caretaker, guard, vendor supervisor; custom roles from the permission list
Approval rules	By amount and type: credit notes, vendor bills, restructures, deposit deductions, remittances
Reminder schedules and message templates	Bills, instalments, rent, deposits, leases, licences
Custom fields	Extra fields on properties, units, parties, leases and work orders, shown in forms, filters and exports
Numbering	Unit codes, contract, lease and document numbers with tenant prefixes

4.8 Plans and gating

Each tenant's plan is held in subscriptions-api. Feature and limit checks (units under management, staff users, modules such as sales, leasing or marketplace publishing) run on every request and in event consumers, following the platform's subscription gating standard. A lapsed subscription keeps data readable but stops new records, never deleting anything.

4.9 Tenant onboarding

1. Organisation created in auth-api (assisted in Release 1, self-service sign-up with trial from Release 2), with plan assigned.
2. Tenant type chosen (estate operator, property manager, landlord), which loads the matching defaults: charge types, document templates, roles.
3. Paybills and bank accounts connected in treasury; ledgers and cost centres created.
4. Properties, units, parties and balances imported from templates, with a dry run.
5. Staff invited and assigned to properties; providers loaded.
6. Go-live checklist completed: live KES 1 payment tests, one parallel billing run, legal templates approved.

SECTION 5

Functional requirements

Priority within a release: **M** must have, **S** should have, **C** could have. Release 1 M and S requirements are in the Shaba Village build price.

ID	Requirement	Rel.	Pri.
PLATFORM AND TENANCY			
FR-01	Every record is scoped to a tenant and protected by row-level security; tenant comes from the signed session.	1	M
FR-02	Tenant settings: profile, branding, custom portal domain, numbering, roles, approval limits, message templates.	1	M
FR-03	Plans and limits enforced per tenant (units, staff users, modules), with upgrade prompts.	1	M
FR-04	Platform console: tenants, plans, usage, support access grants with logging.	1	M
FR-05	Self-service tenant sign-up with trial, tenant type selection and guided set-up.	2	M
FR-06	Seeded charge catalogue; tenants enable, rename and price seeded charges and add custom charges with basis, frequency, scope, bill-to, fund, tax, proration, penalty and priority, without a release.	1	M
FR-07	Layered configuration (platform, tenant, property, unit type, unit) for charges, templates, SLAs, reminders and approval rules; custom fields on key records.	1	M
FR-08	Tenants enable modules and use case presets per tenant and per property; navigation, forms, portals, roles, API routes, jobs, event consumers, notifications and reports include only enabled modules.	1	M
FR-09	Disabling a module hides it and stops its jobs while keeping its data read-only and exportable.	1	M
FR-10	Per-tenant export of all data and documents, and deletion on exit.	1	M
PORTFOLIOS, PROPERTIES AND UNITS			
FR-11	Record properties of any type (estate, apartment block, office, retail, mixed use, industrial, land) with location, photos, amenities and documents.	1	M
FR-12	Record blocks and units with code, use, size, bedrooms, parking, meter serials, entitlement and status.	1	M
FR-13	Assign staff to properties with property-scoped roles (property manager, caretaker, finance, letting); staff see only assigned properties.	1	M
FR-14	Owners, joint owners, occupants, household members, vehicles and domestic staff with dated relationships.	1	M
FR-15	Import properties, units, parties, contracts and opening balances from templates with validation and dry run.	1	M
FR-16	Landlord client portfolios with management mandate: scope, fee basis, expense approval limit, remittance account and day.	2	M
FR-17	Landlord portal: occupancy, rent roll, arrears, expenses, statements, remittances, approvals and documents.	2	M
ACCOUNTS AND SIGN-IN			

FR-18	Customers, landlords and vendor supervisors sign in with mobile number and one-time password; staff use single sign-on with MFA.	1	M
FR-19	Terms and privacy notice accepted per tenant on first sign-in; version and time recorded.	1	M
UNIT SALES AND INSTALMENTS			
FR-20	Price lists by unit type and phase with effective dates; availability by sale status.	1	M
FR-21	Reservations for a set period on payment of a reservation fee; automatic release on expiry.	1	M
FR-22	Sale contracts with buyer KYC, price, discounts, deposit, terms, advocates and the signed agreement.	1	M
FR-23	Instalment schedules (monthly, quarterly, milestone) or outright purchase; restructuring under approval with history kept.	1	M
FR-24	Instalment payments by sales paybill or prompt, oldest due first; reminders; late charges only as the agreement provides; default escalation.	1	M
FR-25	Purchase statement, completion statement, handover checklist with snag list, title stage tracking, defects claims.	1	M
BILLING AND UTILITIES			
FR-26	Charge types, rates, bases (fixed, per type, entitlement, metered, percentage), effective dates, fund and bill-to rules.	1	M
FR-27	Meter rounds with a photo per reading, anomaly flags, estimates, meter replacement, bulk and borehole readings, water balance.	1	M
FR-28	Monthly billing runs per property: one invoice per unit and fund, idempotent on retry.	1	M
FR-29	One-off invoices (levies, recharges, parking) and credit notes under approval.	1	M
FR-30	Bill queries within a set window, handled in a finance queue.	1	S
PAYMENTS AND COLLECTIONS			
FR-31	Paybill payments matched by account number, allocated by configured order, receipted by SMS or WhatsApp.	1	M
FR-32	Prompt payments from the portal; bank transfers and cheques recorded with references.	1	M
FR-33	Suspense queue for unmatched payments; credits held for overpayment.	1	M
FR-34	Escalating reminders, demand letters, payment plans, clearance certificates.	1	M
FR-35	Statements per party and fund on demand in PDF and spreadsheet.	1	M
LEASING AND RENT			
FR-36	Rental applications with consent, identity and income documents, references and approval by manager and, where the mandate says, landlord.	2	M
FR-37	Leases generated from templates: parties, unit, term, rent, escalation, deposit, charges, utilities, notice periods, special conditions; signed and stored.	2	M
FR-38	Deposit invoiced and held in a deposit liability account; refund with itemised deductions on move-out.	2	M
FR-39	Move-in and move-out inspections with room-by-room condition, photos, meter readings, keys and signatures.	2	M
FR-40	Rent invoices on the lease schedule with utilities and service charge; proration on start and end.	2	M

FR-41	Rent reviews and escalations with notice letters; renewals; termination notices with the lease's notice period.	2	M
FR-42	Commercial leases: VAT on rent where applicable, service charge recovery, rent-free periods, turnover clauses, withholding tax by lessee.	2	S
FR-43	Turnover projects between tenancies: repaint, repairs, cleaning, with budget, approval and readiness date.	2	M
FR-44	Landlord remittances: rent collected less management fee, approved expenses and withholding tax, paid on the mandate day with a statement.	2	M
FR-45	Rental income tax support: MRI computation per landlord, withholding where the manager is an appointed agent, PRN generation through treasury.	2	S

DOCUMENTS AND AGREEMENTS

FR-46	Template library per tenant with merge fields, versions and approval; Codevertex starter templates per tenant type.	1	M
FR-47	Generate documents (agreements, forms, notices, letters, certificates) as PDF from templates, stored against the unit and parties.	1	M
FR-48	Signature by recorded acceptance with OTP, or upload of the wet-signed scan; document status from draft to executed.	1	M
FR-49	Document expiry and key-date reminders: lease end, rent review, licences, insurance, title stages.	1	M
FR-50	Advanced electronic signature through an accredited provider.	4	S

SERVICE PROVIDERS AND MAINTENANCE

FR-51	Vendors with KRA PIN, payment details and compliance documents with expiry alerts; vendors may serve several tenants.	1	M
FR-52	Contracts with scope, fee basis, SLAs, service credits and renewal; recurring visit schedules.	1	M
FR-53	Visit evidence with time and photos; missed visits flagged; resident reports.	1	M
FR-54	Security posts, rosters, guard sign-on, patrol checkpoints and coverage reports.	1	M
FR-55	Agency personnel register with badges, deactivated on removal.	1	M
FR-56	Vendor invoices with eTIMS number checked against contract and evidence; approval and payment with withholding tax.	1	M
FR-57	Requests from residents and lessees; work orders with priority, SLA, assignee, quotes, cost, photos, confirmation and recharge.	1	M
FR-58	Preventive maintenance schedules for shared assets.	1	M
FR-59	Landlord approval of repairs above the mandate limit.	2	M

SECURITY AND ACCESS

FR-60	Visitor passes (single, recurring, domestic staff, delivery, contractor) with QR and 6-digit code.	1	M
FR-61	Gate verification, walk-in host approval, entry and exit log, offline operation for 8 hours.	1	M
FR-62	Digital occurrence book and incidents with alerts.	1	M

STAFF, COMMUNICATION AND REPORTING

FR-63	Regular and casual staff, attendance and payroll in the ERP; work orders assignable to ERP staff.	1	M
FR-64	Notices and emergency alerts by property, block or audience across SMS, WhatsApp, email and in-app.	1	M
FR-65	Budgets against actual, income and expenditure, sinking fund statement.	1	M
FR-66	Reports and dashboards per tenant, property and portfolio, exportable.	1	M
FR-67	Data subject requests with deadlines; audit log of all financial, contract and access changes.	1	M
PUBLIC MARKETPLACE			
FR-68	Listings for rent or sale: residential, office, retail, industrial, land; price, size, photos, video link, amenities, location.	3	M
FR-69	Publish a vacant managed unit as a listing in one step; listing closes when a lease or sale is signed.	3	M
FR-70	Lister verification: identity, EARB registration for agents, ownership or authority documents for sale listings.	3	M
FR-71	Moderation: photo and price checks, duplicate detection, reports from the public, takedown with reasons.	3	M
FR-72	Search with filters, map view, sort, share links; listing pages rendered for search engines.	3	M
FR-73	Enquiries with masked contacts, delivered to the lister and their CRM; viewing booking.	3	M
FR-74	Online rental applications into the lister's leasing workflow; offers on sale listings into sales.	3	S
FR-75	Saved searches and alerts; favourites.	3	S
FR-76	Listing plans and featured placement paid through treasury.	3	S

SECTION 6

Non-functional requirements

ID	Attribute	Requirement
NFR-01	Tenant isolation	No request, query, file link, event or report can return another tenant's data; enforced in the database by row-level security and covered by automated cross-tenant tests on every release.
NFR-02	Configurability	Charges, catalogue entries, templates, modules and use cases are configured by tenants without code changes or releases; no screen, route, job or message of a disabled module is reachable.
NFR-03	Mobile performance	Portal initial transfer under 200 KB excluding images; Largest Contentful Paint under 2.5 seconds on a mid-range Android phone over 3G; images as resized WebP.
NFR-04	Response time	95th percentile under 300 ms for reads and 600 ms for writes, excluding M-Pesa latency; gate verification under 1 second online or offline; marketplace search under 500 ms.
NFR-05	Capacity and growth	Designed for 500 tenants, 200,000 units, 2 million gate events and 300,000 payments a month, scalable tenfold without redesign: keyset pagination, monthly partitions for event tables, daily aggregates, read replicas.
NFR-06	Batch performance	A billing run for 1,000 units completes within 2 minutes and is safe to retry; runs for different tenants proceed in parallel under per-tenant limits.
NFR-07	Availability	99.5% monthly for portals, gate service and payment matching; 99.9% target for the public marketplace from Release 3. Two replicas per service; zero-downtime deployments.
NFR-08	Financial integrity	Each M-Pesa receipt recorded once; each invoice issued once per unit and period; balanced double-entry ledger; daily paybill reconciliation; deposits held as liabilities.
NFR-09	Security	OWASP ASVS Level 2, TLS throughout, field encryption for identity numbers and KRA PINs, MFA for staff, audit trail on financial, contract and configuration changes.
NFR-10	Privacy	Data minimisation, consent records, retention enforced by automated purge per tenant, masked contacts on the marketplace, documented hosting and transfer safeguards.
NFR-11	Accessibility	WCAG 2.2 Level AA.
NFR-12	Search visibility	Marketplace pages server-rendered with structured data, canonical links and sitemaps (Release 3).
NFR-13	Localisation	English at launch; text externalised for Kiswahili; KES with multi-currency display for diaspora users.
NFR-14	Recovery	Daily encrypted offsite backups, 7 daily and 4 weekly copies; quarterly restore drills; RPO 24 hours, RTO 4 hours.
NFR-15	Observability	Structured logs, tracing and alerts per tenant: failed billing runs, unmatched payment build-up, offline gate devices, errors, latency.
NFR-16	Maintainability	Codevertex stack and conventions, automated tests, documented APIs, versioned migrations, configuration over custom code.

SECTION 7

Solution architecture

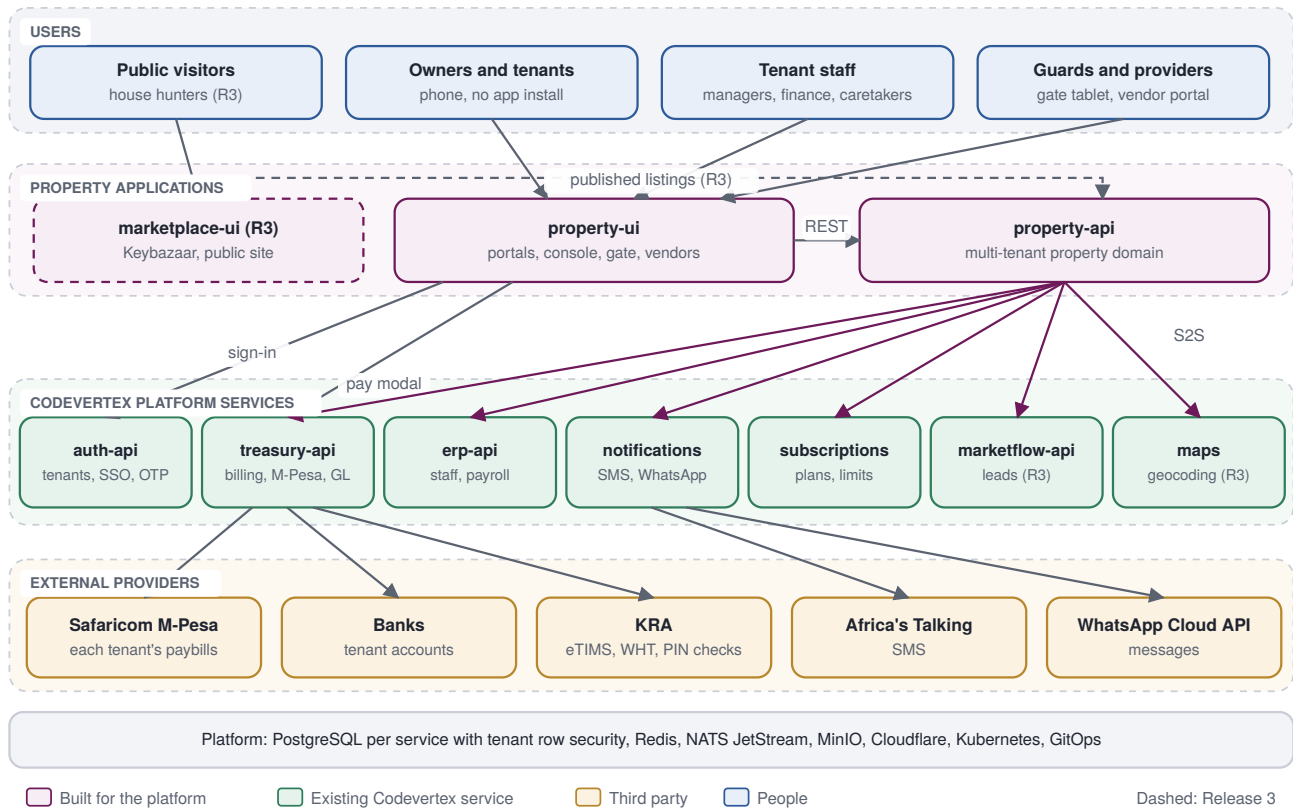


Figure 3. Solution architecture

7.1 Components

Component	Status	Function
property-ui	New (R1)	Next.js progressive web application with tenant-branded surfaces: owner, lessee and landlord portals, management console, gate tablet and vendor portal. Uses the shared UI library for sign-in, the payment modal, data tables with export, document preview, legal links and cookie consent.
property-api	New (R1)	Go service for the multi-tenant property domain: portfolios, properties, units, parties, charge rules, meters, billing runs, sale contracts, leases, inspections, documents and templates, work orders, vendor contracts and visits, posts, passes, gate events, incidents, listings and statistics. Does not move money.
marketplace-ui	New (R3)	Keybazaar: the public, server-rendered site for search, listing pages, map, enquiries, viewings and applications, reading a published listing projection from property-api.
auth-api	Existing, extended	Tenants, branches, staff single sign-on with MFA, roles, sessions, terms acceptance, API keys. Extended with mobile one-time password sign-in for customers.
treasury-api	Existing, extended	Customer accounts, invoices, credit notes, recurring billing, payment intents, M-Pesa prompt and paybill, allocation, dunning, statements, vendor bills, approvals, payouts, withholding tax and rental tax PRNs, eTIMS, budgets, ledger. Extended with paybill matching by unit code, instalment plans linked to sale contracts, deposit liability handling and landlord remittance runs.

erp-api	Existing	Staff records, contracts, attendance, leave, payroll with statutory deductions, casual worker payments, asset register.
notifications-api	Existing	SMS, WhatsApp, email and push with per-tenant sender, templates and delivery status.
subscriptions-api	Existing	Plans, feature entitlements and limits per tenant.
marketflow-api	Existing (R3)	CRM leads and contacts for marketplace enquiries and buyer pipelines.
maps	Existing (R3)	Geocoding and map tiles for property locations and map search.
Shared libraries	Existing	Authentication client, service client, event outbox, cache with job leases, rate limiting, pagination, HTTP middleware, shared UI library.

7.2 Architecture principles

- **Tenant first.** Tenant ID on every table, row, event, cache key, file path and job; isolation tested on every release.
- **Single data owner.** property-api stores treasury account, invoice, plan and vendor IDs, ERP employee IDs and auth user IDs, and reads balances from treasury.
- **Single payment path.** Only treasury-api talks to M-Pesa and banks.
- **Event driven.** Payment, invoice and payout events reach property-api through the transactional outbox and NATS; consumers are idempotent by event ID.
- **Replica safe.** Billing runs, reminders, SLA timers, remittances and purges run once per period across replicas using shared leases.
- **Public and private separated.** The marketplace reads a published projection of listings; it has no route to tenant operational data.
- **Standard stack.** Go with Ent and Atlas migrations, Next.js, Helm and ArgoCD.

7.3 What already exists

Need	Existing capability used
Organisations, staff, branches, roles, API keys	auth-api tenants and branches
Monthly invoices with numbering, tax and due dates	treasury invoicing and recurring schedules
M-Pesa prompts and paybill confirmations recorded once	treasury M-Pesa gateway and C2B inbox
Escalating overdue reminders with a pay link	treasury dunning worker
Balances, ageing and statements	treasury receivables and payables
Vendor bills, approval with OTP, payment by bank or M-Pesa	treasury vendor bills, approvals engine, payout dispatcher
Withholding tax, rental tax PRNs, eTIMS	treasury tax module and KRA integrations
Instalment plans	treasury instalment plans, extended for sale contracts
Payroll, casual workers, attendance, assets	erp-api
SMS, WhatsApp and email	notifications-api
Leads and pipelines	marketflow-api

SECTION 8

Estate management: Shaba Village configuration

How the estate management module is configured for the launch tenant: who pays what, into which account, and how Shaba Village's two money streams stay apart. Other estate operators use the same module with their own catalogue choices and rules.

8.1 Charges

Charge	Billed to	Basis	Fund
Unit price	Buyer	Outright, or deposit plus instalments per the sale agreement	Sales
Reservation fee	Prospective buyer	Fixed, credited to the deposit when the agreement is signed	Sales
Service charge	Owner	Fixed per unit type, or annual budget divided by unit entitlement	Estate
Water	Owner, or the occupant if the owner assigns it	Metered: consumption times the estate tariff, plus any fixed meter charge	Estate
Garbage collection	Owner or occupant	Fixed per unit, or included in service charge	Estate
Security levy	Owner	Fixed per unit, or included in service charge	Estate
Sinking fund	Owner	Fixed or percentage of service charge, kept in its own ledger account	Estate (reserve)
Extra parking	Owner or occupant	Per additional bay	Estate
Special levy	Owner	One-off amount approved at a general meeting, by entitlement	Estate
Recharges	Owner	Cost of in-unit repairs or damage, with the work order as evidence	Estate
Late payment charge	Owner or buyer	Only where the by-laws or sale agreement provide; capped and never compounded	As the debt

These charges are Shaba Village's selection from the platform charge catalogue, with Shaba's names and rates; any charge can be added, renamed or retired by Shaba's finance team without a release. A rate change applies from its effective date and never alters invoices already issued.

8.2 Worked example: monthly bill

Unit B07, three bedroom, October 2026	KES
Service charge, three bedroom rate	4,500
Water: reading 1,284 less 1,275 = 9 m3 at KES 150	1,350
Garbage collection	300

Sinking fund	300
Total due by 10 November 2026	6,450

Rates are illustrative and are set by Shaba Village. For comparison, the Mavoko Water and Sewerage Company approved domestic tariff ranges from KES 120 per m3 for the first 6 m3 to KES 170 per m3 above 300 m3.

8.3 Who receives the bill

The owner is liable for all estate charges by default. Where an owner lets the unit, the owner may assign water, garbage and parking to the tenant, who then receives those bills and pays with the same unit account number. Service charge, sinking fund and special levies remain with the owner. If an assigned tenant leaves with an unpaid balance, the balance reverts to the owner after 30 days, as the owner terms state.

8.4 Fund segregation

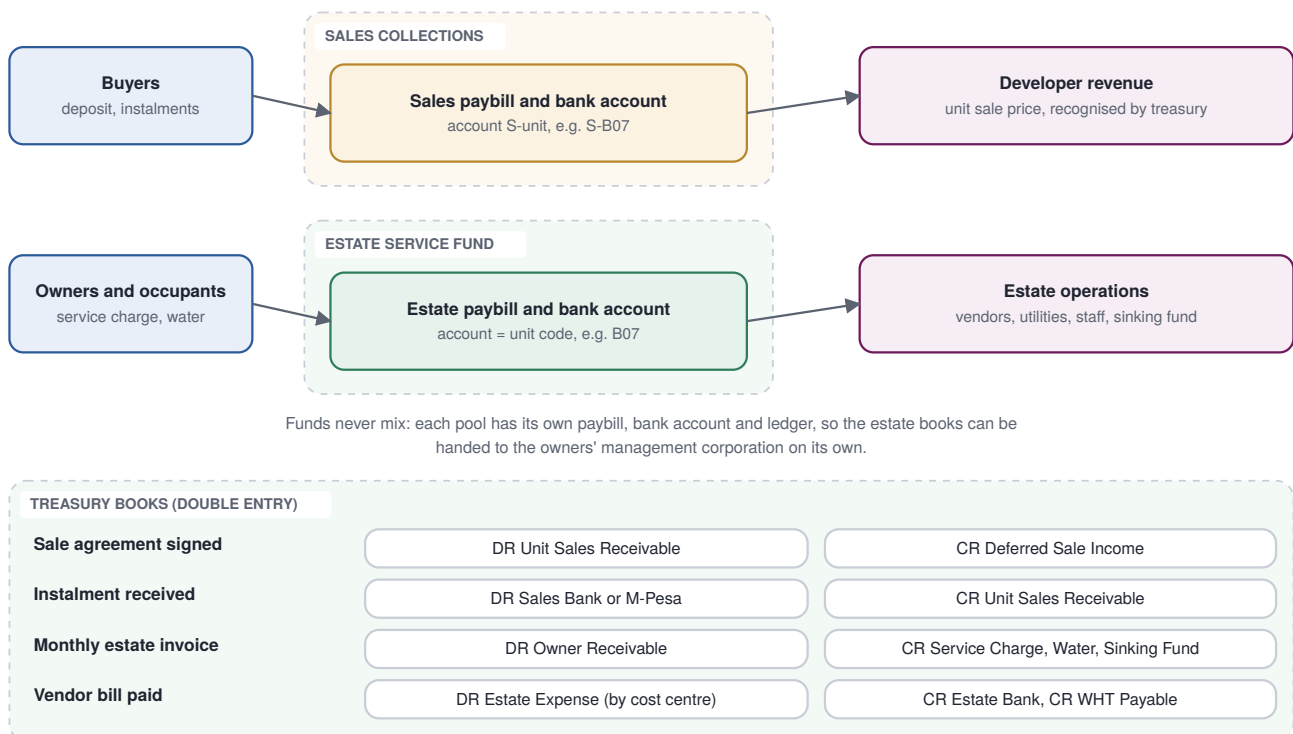


Figure 4. Fund segregation and accounting

Sale proceeds belong to the developer. Estate charges are collected to run the estate on behalf of all owners. Mixing the two is the most common cause of service charge disputes in Kenyan estates. The platform keeps them apart by design: separate paybills and bank accounts, separate cost centres and ledger accounts in treasury, and separate statements. A buyer who is also an owner sees both balances, but a payment to one account can never settle the other.

8.5 Payment references

Payment	Paybill	Account number
Estate charges	Estate paybill	Unit code, for example B07
Unit purchase	Sales paybill	S plus unit code, for example S-B07

Owners may also pay by M-Pesa prompt from the portal, which needs no reference, or by bank transfer with the unit code as reference. Payments that cannot be matched are placed in a suspense queue for finance, with the payer's name and number to assist.

8.6 Transition to the management corporation

Under the Sectional Properties Act, 2020, common property in a sectionally registered estate is owned and managed by a management corporation of the unit owners. The estate fund books, unit accounts, sinking fund, vendor contracts and records are kept so that they can be handed over complete when the corporation is formed. Shaba Village can then continue as managing agent for the corporation on the same platform, or hand over administrative control entirely.

SECTION 9

Unit sales and instalments

Every unit sale, from reservation to title, is tracked against the unit, with the buyer able to see exactly what has been paid and what is next.

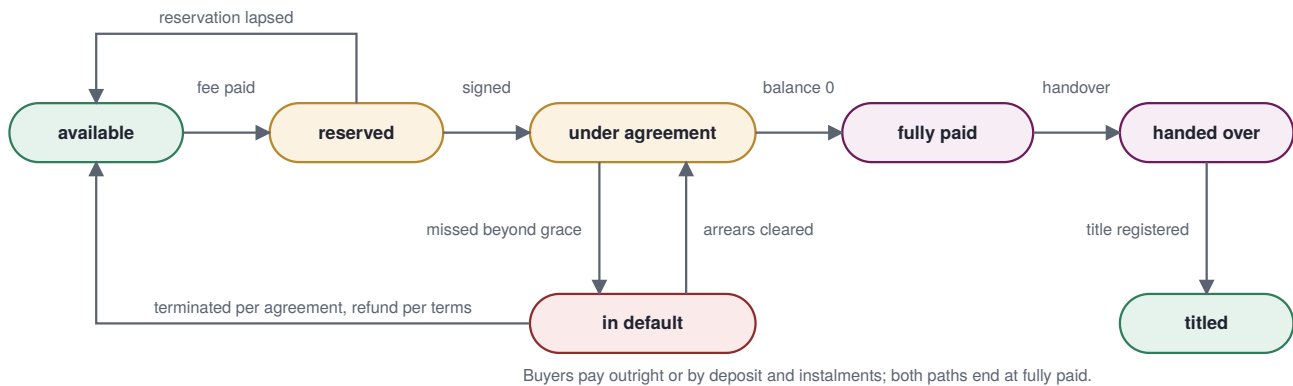


Figure 5. Unit sale lifecycle

9.1 Payment options

Option	How it is recorded
Outright purchase	One invoice for the balance after the reservation fee, due on the agreed date. Part payments are accepted and shown on the statement.
Deposit and monthly instalments	Deposit invoice, then a schedule of equal instalments over the agreed term. The last instalment absorbs rounding.
Milestone instalments	For units bought before completion: instalments fall due on agreed construction milestones, each released by the sales officer with evidence.
Mortgage or SACCO financing	Deposit from the buyer; the financier's payment recorded on release, referenced to the financier's letter of offer.

9.2 Worked example

Item	KES
Three bedroom apartment, price	7,500,000
Reservation fee, credited to deposit	100,000
Deposit at 20%, including the reservation fee	1,500,000
Balance over 24 monthly instalments	250,000 each
Total	7,500,000

Figures are illustrative. Prices, deposits, terms and any interest are set by Shaba Village in each sale agreement.

9.3 Rules

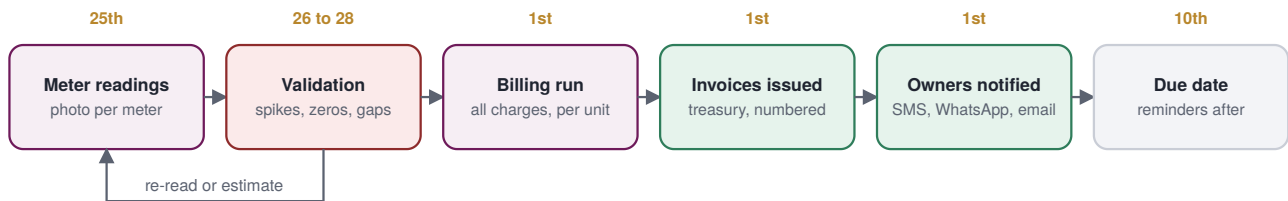
Situation	Rule
Payment received	Applied to the oldest due instalment, then the next. An overpayment reduces the following instalment unless the buyer asks for early completion.
Missed instalment	Reminders at 3 days before, on the due date, and 7 and 14 days after. A late charge applies only if the agreement provides one, at the agreed rate, never compounded.
Default	After the grace period in the agreement, the contract moves to default and the sales officer is alerted with the full history. Notices and any termination are issued by the advocate under the agreement; the platform records each step.
Restructuring	A new schedule may be agreed. It needs approval, keeps the original schedule on record and is shown to the buyer for acceptance.
Termination and refund	Refund amounts follow the agreement and are paid through treasury under approval. The unit returns to available only after the refund is recorded.
Fully paid	A completion statement and receipt are issued, and the handover checklist opens.
Handover	Meter readings, keys, snag list and signatures recorded. Estate billing starts from that date, prorated for the first month.
Title	Each title stage is logged with date and document until the title is released to the owner.

Buyer identity documents and KRA PINs are needed for the sale agreement and title transfer. They are encrypted, visible only to the sales role, and kept for the statutory retention period.

SECTION 10

Billing, utilities and collections

10.1 Monthly cycle



A billing run is keyed by estate and month, so a retry never issues a second invoice for the same unit and period.

Figure 6. Monthly billing cycle

Dates are configurable. The billing run asks treasury to create one invoice per unit and fund with the month's line items. Each invoice carries the unit code as its payment reference and a link to pay. Owners receive a short message with the amount, due date and paybill details, and the full bill in the portal.

10.2 Meter readings

1. The caretaker opens the reading round on a phone. Units are listed in walking order by block.
2. Each reading is typed and photographed. The photo is stored with the reading and shown on the bill.
3. The platform flags readings lower than the previous one, zero consumption for an occupied unit, and consumption above three times the unit's 3-month average.
4. Flagged readings are re-checked. A meter that cannot be read is billed at the 3-month average, marked estimated, and corrected on the next actual reading.
5. A meter replacement records the closing reading of the old meter and the opening reading of the new one, so no consumption is lost or billed twice.

10.3 Water balance

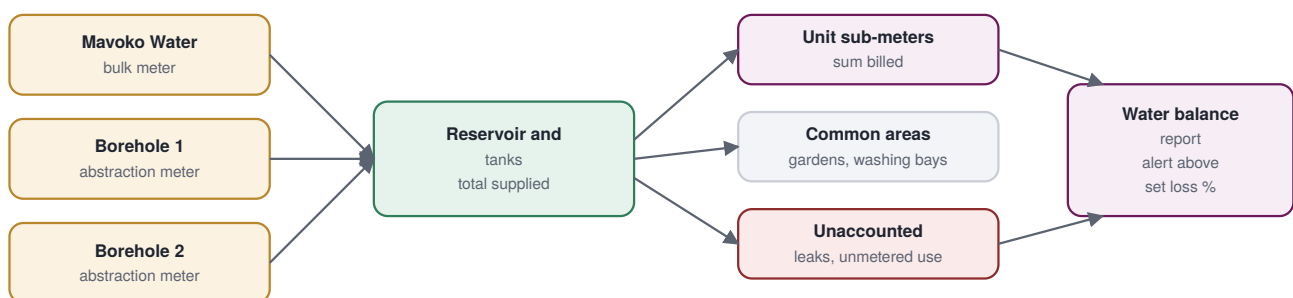


Figure 7. Water balance

Bulk readings from the Mavoko connection and each borehole meter are entered monthly. The report compares total water supplied with water billed to units and estimated common area use. The difference is unaccounted water, shown as a percentage and trended by month. Above the set threshold, for example 15%, the estate manager is alerted to check for leaks.

10.4 Payment matching and allocation

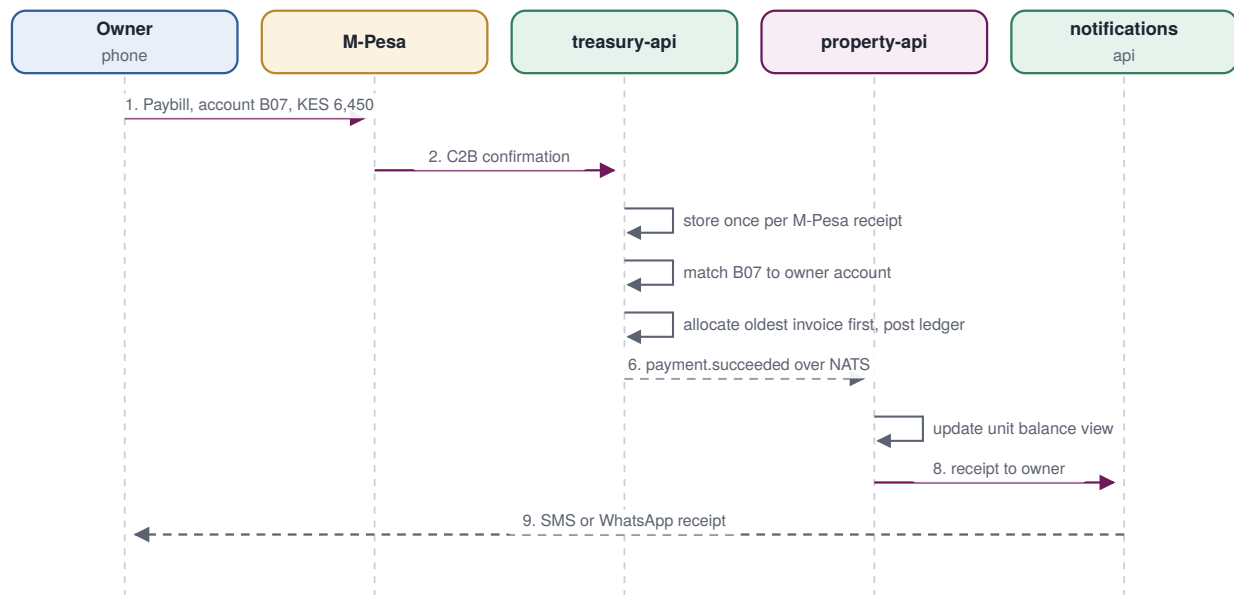


Figure 8. Paybill payment sequence

- Each M-Pesa confirmation is stored once under its receipt number, so a repeated callback never creates a second payment.
- The account number is normalised (case, spaces, dashes) and matched to the unit. Close misses, such as b 07 or B7, are matched; anything else goes to the suspense queue.
- Allocation follows the configured order: oldest invoice first by default, or charge type priority if the by-laws require it. Late charges are settled last.
- Daily, treasury compares the paybill statement with recorded payments. Differences are listed for finance.

10.5 Arrears

Days after due date	Action
1	Friendly reminder with balance and pay link AUTO
7	Second reminder AUTO
14	Firm reminder; late charge applied if the by-laws provide AUTO
30	Call list for the finance officer; offer of a payment plan
45	Demand letter generated for review and sending
60 and above	Escalation to management; any measures permitted by the by-laws, such as withholding a clearance certificate

Not automated: disconnection of water or any other supply, locking of gates or denial of entry to a resident for arrears. These require due process and are excluded from the platform by design.

10.6 Budget and transparency

The annual service charge budget is entered in treasury by cost line: security, cleaning, garbage, landscaping, water production, common power, repairs, management, insurance and the sinking fund. Each vendor bill and payroll cost is posted to the matching line. Owners can see a monthly summary of budget against actual, and the income and expenditure statement is produced for the AGM directly from the ledger.

SECTION 11

Rental and portfolio management

Release 2 opens the platform to property management firms and landlords. It reuses the Release 1 property, billing, works and document engines, and adds leases, deposits and landlord remittances.

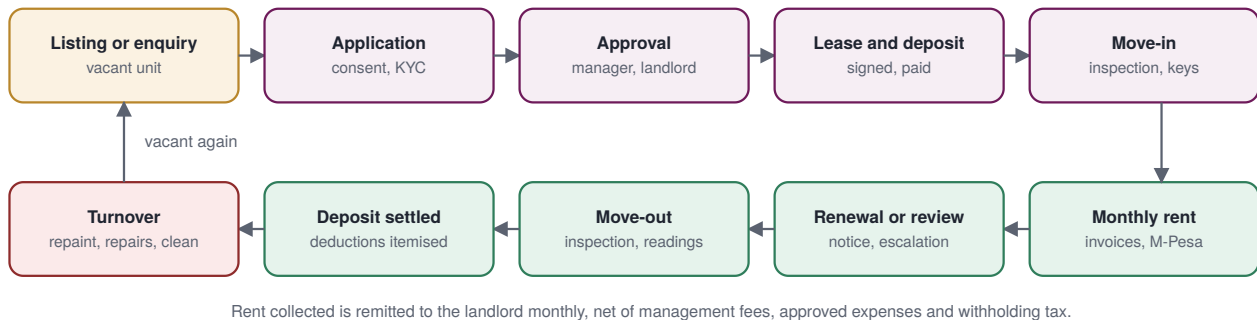


Figure 9. Lease lifecycle

11.1 Landlord clients and mandates

Item	What is recorded
Landlord	Individual or company, KRA PIN, residency status for tax, contacts, bank or M-Pesa remittance account verified by OTP
Management mandate	Properties covered, services (letting only, rent collection, full management), start and end, notice period, signed agreement
Fees	Management fee as a percentage of rent collected or fixed per unit; letting fee per new lease; renewal fee; fees on repairs; all as configurable charges against the landlord
Expense authority	Repairs the manager may approve alone, and the limit above which the landlord approves in the portal
Remittance	Day of month, minimum float retained for repairs, statement format, tax withholding where the manager is an appointed agent

11.2 Property staff

Each property has its team: a property manager, one or more caretakers, and finance and letting officers as needed, all assigned in the console. Staff see the properties assigned to them; a caretaker's phone shows only their property's readings, requests, inspections and notices. Firm-employed staff are paid through the ERP; landlord-paid caretakers can be recorded as a property expense instead.

11.3 Tenant onboarding

1. A vacant unit is advertised, from the console or (Release 3) on the marketplace, or an applicant is entered by the letting officer.
2. The applicant applies on their phone: identity, contacts, employer or business details, income evidence, references and next of kin, with consent to the checks described in the privacy notice.
3. The letting officer reviews; where the mandate requires, the landlord approves in the portal.
4. The lease is generated from the tenant's template with the agreed terms, sent for acceptance by OTP or signature, and stored once executed.
5. The first invoice for deposit, first rent and any move-in fee is issued; on payment the lease activates.
6. The move-in inspection records each room's condition with photos, meter readings and keys handed over, signed by both sides. The lessee gets portal access, gate passes and the house rules.

11.4 Rent and charges

- Rent and recurring charges invoice on the lease schedule (monthly, quarterly or as agreed) with utilities from meter readings and service charge from the catalogue.
- Each lease has its own paybill account reference, so rent payments match automatically, as for owners in Release 1.
- Escalations apply on the lease's review dates (fixed percentage or agreed amount), with a notice letter generated in advance as the lease requires.
- Commercial leases carry VAT on rent where the landlord is registered, service charge recovery, rent-free periods and lessee withholding tax on rent, with the withholding certificate captured against the payment.

11.5 Deposits

Deposits are liabilities, not income. Each deposit is held in a deposit ledger against the lease, shown on the lessee's and landlord's statements, and never counted in rent remitted. On move-out, deductions for unpaid rent, utilities and damage beyond fair wear and tear are itemised with the inspection photos, approved, and the balance refunded to the lessee within the period in the lease, typically 30 days.

11.6 Renewals, notices and move-out

Event	Platform action
Lease end approaching	Reminders at 90, 60 and 30 days to manager and landlord; renewal offer generated
Rent review	Review notice generated with the lease's notice period; the new rent applies from the effective date
Notice to vacate	Recorded from either side with date and reason; move-out date set; the unit is marked as coming vacant
Move-out	Final meter readings, inspection against the move-in report, keys returned, final invoice and deposit settlement
Arrears	Reminders, demand letters and payment plans; legal steps by advocates. No lock-outs, removal of doors or disconnection.

11.7 Turnover: repaint and repairs

When a unit is coming vacant, a turnover project opens with tasks from the move-out inspection: repainting, plumbing and electrical repairs, deep cleaning, pest control and replacement items. Each task is a work order with a quote, cost

and photos. Costs above the mandate limit go to the landlord for approval. The project sets a ready date, which drives the listing's availability date.

11.8 Landlord statements and remittances

Statement line	Source
Rent and charges invoiced, collected and outstanding by unit	treasury invoices and payments
Less management and letting fees	Mandate fee rules
Less approved repairs and property expenses	Vendor bills and work orders
Less tax withheld, where applicable	MRI at 7.5% for resident landlords when the manager is an appointed withholding agent; non-resident withholding at 30%
Net remitted	Treasury payout to the landlord's verified account, under approval

Rent collected for landlords is held in the manager's client account ledger, separate from the manager's own income, and every shilling is traceable from paybill to remittance.

SECTION 12

Documents and agreements

Every agreement, form, notice and certificate in the property lifecycle is generated from a template, stored against the unit and parties, tracked to execution and kept for its retention period.

12.1 Document catalogue

Document	Parties	Generated from	Rel.
SALES			
Reservation form and receipt	Developer, buyer	Reservation	1
Letter of offer	Developer, buyer	Price list and terms	1
Sale agreement (draft for advocates) and payment schedule annex	Developer, buyer	Sale contract, instalment plan	1
Instalment reminder, default and restructure letters	Developer, buyer	Instalment plan	1
Completion statement and certificate of full payment	Developer, buyer	Sale contract	1
Handover form, snag list, defects notice	Developer, owner, contractor	Handover checklist	1
Title stage records: sectional plan, lease or title, consents, stamp duty receipts	Advocates, Land Registry	Uploaded	1
ESTATE AND OWNERS			
Owner welcome pack, by-laws, house rules	Estate, owner	Template and library	1
Bill-to assignment consent (owner to occupant)	Owner, occupant	Occupancy	1
Demand letter, payment plan agreement	Estate, owner	Arrears	1
Clearance certificate for resale	Estate, owner	Account balances	1
AGM notice, budget, income and expenditure, sinking fund statement, minutes	Estate, owners	Budgets and ledger	1
LETTINGS			
Management agreement (mandate)	Manager, landlord	Mandate	2
Rental application and consent to checks	Applicant	Application	2
Residential tenancy agreement	Landlord or manager, lessee	Lease	2
Commercial lease or letter of offer to lease	Landlord, business lessee	Lease	2
Guarantor agreement	Guarantor	Lease	2
Move-in and move-out inspection reports	Manager, lessee	Inspection	2
Rent review, renewal offer, notice to vacate, termination notice	Landlord, lessee	Lease	2

Deposit settlement statement	Manager, lessee	Move-out	2
Landlord statement and remittance advice	Manager, landlord	Remittance run	2

PROVIDERS AND STAFF

Service contract with SLA schedule and confidentiality clause	Tenant, vendor	Vendor contract	1
Work order, quote approval, completion certificate	Tenant, vendor	Work order	1
Remittance advice and withholding tax certificate	Tenant, vendor	Payment	1
Employment contracts and payslips	Tenant, staff	ERP	1

PLATFORM

Privacy notice, portal terms, gate notice, data processing agreement	Tenant, users	Legal pack	1
Lister terms, listing verification documents	Lister, platform	Listing	3

12.2 Templates

- Codevertex seeds starter templates for each tenant type. Each tenant adapts them, or uploads its advocate's own wording, before use. Templates are versioned and must be approved by the tenant administrator before they can be issued.
- Merge fields pull from the record: parties, unit, property, prices, schedules, dates, meter readings, balances. Conditional clauses switch on configuration, for example a guarantor clause only when a guarantor exists.
- Generated documents are PDFs in the tenant's branding with a reference number and a verification code, so a copy can be checked against the original.

12.3 Execution and signatures

Method	Use	Rel.
Acceptance by OTP	Portal terms, bill-to consent, payment plans, inspection reports, renewal offers: the signer reviews and accepts with a one-time password; name, time, document hash and device are recorded	1
Wet signature and upload	Sale agreements, leases over two years, mandates and documents for registration: signed in person, scanned and uploaded against the record	1
Advanced electronic signature	Through an accredited provider for land-related contracts, which the Law of Contract Act has recognised since 2020	4

Each document moves through draft, issued, partly signed, executed, and expired or superseded. Key dates in documents (lease end, review dates, notice deadlines, licence expiry) become reminders. Leases over two years must be registered under the Land Registration Act, and leases attract stamp duty under the Stamp Duty Act; the platform flags both on generation and records the registration and stamp duty references when the advocate returns them.

12.4 Document storage and access

- Stored in object storage under the tenant, linked to the unit, parties and contract, never public; opened through signed links that expire.
- Access by role and relationship: a lessee sees their own lease and inspection reports; a landlord sees the documents of their portfolio; identity documents are visible only to the sales or letting role.

- Every view and download of a contract or identity document is logged.
- Retention follows the schedule in [Legal and regulatory compliance](#); documents under a legal hold are not purged.

SECTION 13

Public marketplace: Keybazaar

Release 3 launches Keybazaar, which publishes property to the public: homes, offices, shops, warehouses and land, for rent or sale, from tenants on the platform and from verified independent listers.

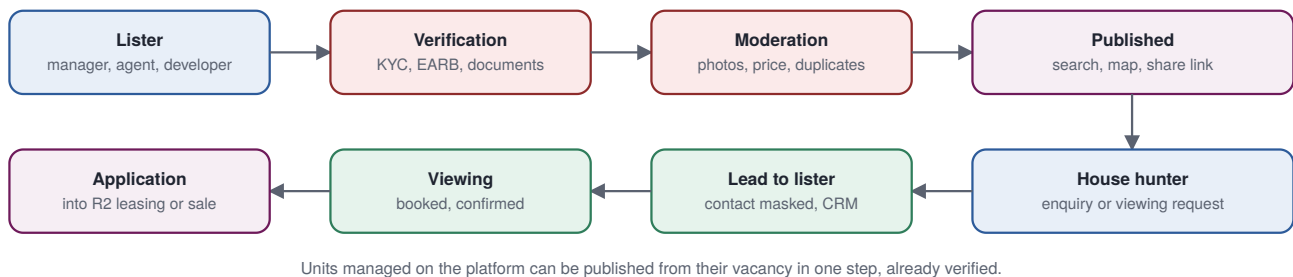


Figure 10. Listing and enquiry flow

13.1 What can be listed

Category	For rent	For sale
Residential	Apartments, houses, bedsitters, serviced units	Apartments, houses, off-plan units
Commercial	Offices, shops, showrooms, co-working space	Office floors, commercial buildings
Industrial	Warehouses, godowns, yards	Warehouses, industrial plots
Land	Leasehold plots, agricultural land	Residential, commercial and agricultural plots

13.2 Sources of listings

- **Managed vacancies.** A vacant or coming-vacant unit on the platform is published in one step with its photos, rent, charges and availability date. It closes automatically when a lease or sale is signed, so stale listings cannot linger.
- **Developers.** Units for sale, including off-plan, from the price list with live availability; enquiries flow into the sales module.
- **Verified independent listers.** Agents and private owners without a full subscription list under a listing plan after verification.

13.3 Trust and verification

Control	How it works
Lister verification	Identity by OTP and document check; agents must give an EARB registration number, checked before publishing; companies give registration and KRA PIN
Authority to list	Sale listings require a title or official search, or the owner's written authority; rental listings by agents require the landlord's authority
Verified badge	Shown for listings from managed properties and verified listers, with the verification date
Moderation	New listings checked for duplicate photos, prices far outside the area range, and contact details hidden in text; public reports with fast takedown

Safe payment rule	Seekers are told never to pay before viewing and signing; deposits and first rent for platform-managed units are paid only to the tenant's verified paybill through the platform
Masked contacts	Seekers and listers talk through the platform until the seeker chooses to share contact details

13.4 Search and discovery

- Filters: purpose, category, location, price range, bedrooms, size, amenities, furnished, parking, availability date, verified only.
- Map search with clustered pins and drawn areas, using the Codevertex maps service; distance to roads, schools and hospitals shown where known.
- Saved searches with alerts by SMS, WhatsApp or email; favourites; share links with preview images.
- Server-rendered listing and area pages for search engines, with structured data.

13.5 Enquiries, viewings and applications

1. A seeker sends an enquiry or requests a viewing slot from the lister's calendar.
2. The enquiry reaches the lister's inbox and, if they use it, their CRM as a lead with its source listing.
3. Viewings are confirmed, reminded and marked attended or missed.
4. For managed units, the seeker applies online and the application enters the tenant's leasing workflow; for sale units, an offer enters the sales module.

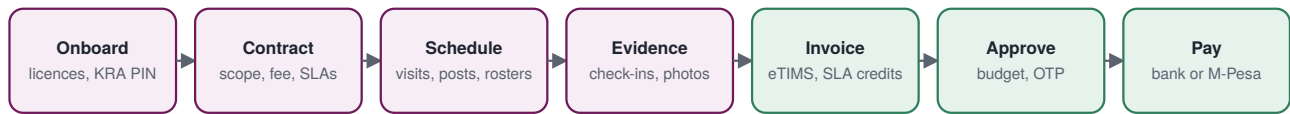
13.6 Monetisation

Revenue line	Basis
Listing inclusion for subscribed tenants	Included in Growth and Professional plans up to a listing limit
Independent listing plans	Monthly plans for agents and private owners by number of active listings
Featured placement	Paid boosts by area and period, through treasury
Developer project pages	Showcase pages for estate launches

SECTION 14

Service providers and property operations

Security, garbage, cleaning and maintenance are bought from outside providers. The platform makes each contract measurable and pays only for service delivered.



property-api runs the first four steps. treasury-api holds the vendor bill, the approval and the payment.

Figure 11. Service provider cycle

14.1 Provider types

Provider	Required documents	Schedule and evidence	Typical SLA
Security agency	PSRA licence, business permit, KRA PIN, insurance	Posts and shifts, roster, guard sign-on, patrol scans, incidents	Every post manned 24 hours; patrol rounds hourly at night; incident report within 1 hour
Garbage collector	NEMA waste transport licence, county permit, KRA PIN	Collection days, check-in photo at each collection point	Collection on scheduled days by 10:00; segregated collection
Cleaning agency	Business permit, KRA PIN, insurance	Daily or weekly rosters by block, checklist with photos	Checklist complete by 09:00; complaints resolved within 24 hours
Landscaping	Business permit, KRA PIN	Weekly visits by zone	Visit completed in its week
Borehole and pump service	Business permit, KRA PIN; WRA driller registration for borehole works	Quarterly service, call-outs as work orders	Breakdown response within 4 hours
Plumbing, electrical, fence, gate	NCA or EPRA licence as applicable, KRA PIN	Work orders	By work order priority
Pest control	PCPB licence, KRA PIN	Quarterly treatment	Visit completed in its month
Tank and reservoir cleaning, water testing	Business permit, accredited laboratory for tests	Twice yearly cleaning; test results uploaded	Results within limits

14.2 Security agency operations

1. Posts are defined with the number of guards per shift, for example main gate two by day and two by night, and a night patrol.
2. The agency supervisor uploads the weekly roster through the vendor portal. Only guards registered for the estate can be rostered.
3. Guards sign on and off at the gate tablet with their badge and PIN. The platform compares sign-ons with the roster and contract and reports unmanned post time.
4. Patrol rounds are scheduled; guards scan QR tags at each checkpoint. Missed checkpoints are reported in the morning summary.
5. Incidents and handover notes are written in the digital occurrence book, visible to the estate manager in real time.
6. At month end, the coverage report is attached to the agency's invoice with any service credit due under the contract.

14.3 Agency personnel

The agency is the employer of its guards and cleaners and remains responsible for their vetting, pay and statutory obligations. Shaba Village records only what it needs to control access: name, agency, role, phone number and badge. A photo is stored only if the agency provides it for identification at the gate. When a person leaves the estate deployment, the badge is deactivated the same day.

14.4 Maintenance work orders

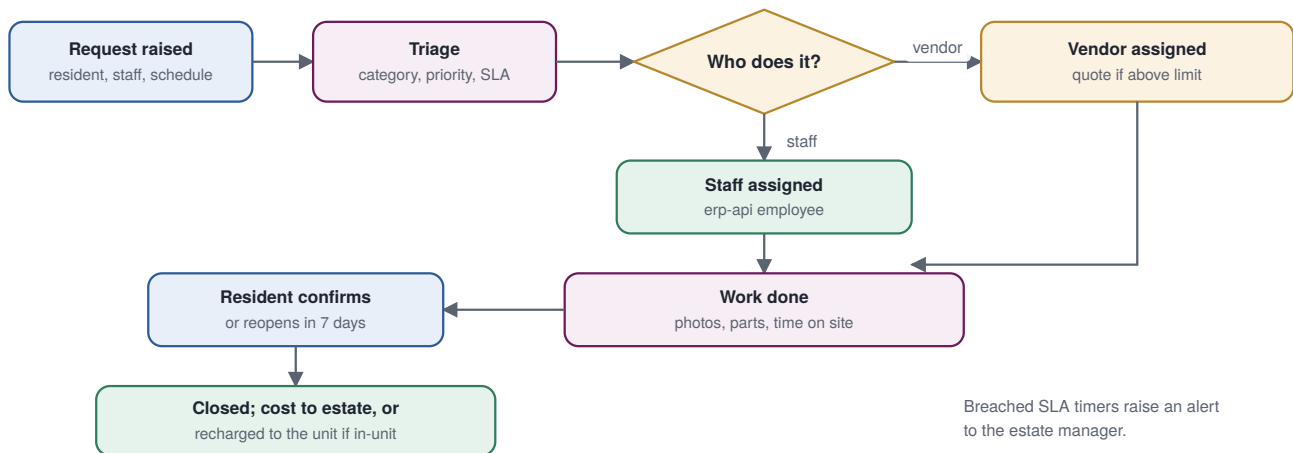


Figure 12. Work order flow

Priority	Examples	Response	Resolution
Emergency	Burst main, sewer overflow, gate or fence down, no water estate-wide	1 hour	4 hours
High	Pump failure, block without water, security light failure	4 hours	24 hours
Normal	Leaking common tap, drainage blockage	24 hours	72 hours
Low	Painting, minor landscaping	3 days	14 days

Repairs inside a unit are the owner's responsibility under most by-laws. Where the estate carries one out at the owner's request, or to stop damage to other units, the cost is recharged to the unit with the work order as evidence.

14.5 Vendor invoices and payment

1. The vendor submits the monthly invoice in the vendor portal with its eTIMS invoice number and document.
2. The platform places the invoice beside the contract fee, completed and missed visits, post coverage and resulting service credit.
3. The estate manager confirms the service; treasury records the vendor bill against the budget line.
4. Approval follows the payout policy: below the limit it is approved by finance, above it by a director with OTP, bound to amount and payee.
5. Treasury pays from the estate account by bank transfer or M-Pesa, deducts withholding tax where it applies, and sends the remittance advice.

14.6 Estate staff

Shaba Village's own regular and casual staff, such as the caretaker, plumber, gardeners and day labourers, are managed in the ERP. Regular staff are paid through monthly payroll with PAYE, NSSF, SHIF and the affordable housing levy. Casual workers are paid per day or per task from the casual payments module, by M-Pesa, with the payment recorded against the work order's cost line. Payroll and casual payments post to the estate fund ledger in treasury. property-api only stores the ERP employee ID on work orders and service checks.

SECTION 15

Security, access and visitors

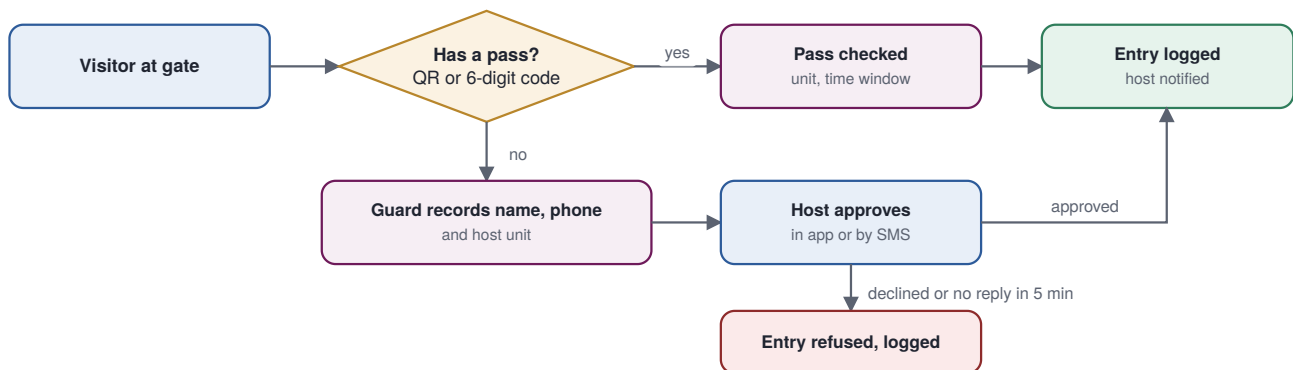


Figure 13. Visitor entry

15.1 Pass types

Pass	Created by	Validity
Guest, single visit	Resident	One entry within the chosen window, default 12 hours
Guest, recurring	Resident	Set days and hours, up to 90 days
Domestic staff	Owner	Set days and hours, renewable, cancellable at any time
Delivery	Resident, or guard with host approval	Single entry, 2 hours
Contractor	Estate manager from a work order	Until the work order closes
Agency personnel	Estate manager	While deployed

15.2 Data collected at the gate

The gate collects the minimum needed for security: the visitor's name, phone number, host unit, time in and out, and vehicle plate if driving. Identity cards are not photographed, copied or retained. Where the by-laws require an identity check, the guard views the card and records only that it was seen. A privacy notice is displayed at each gate and linked in every pass message. Gate events are kept for 90 days and incident records for the period set in the retention schedule.

15.3 Offline operation

Each gate tablet keeps a cache of passes valid for the next 24 hours and of active badges. During a network outage, passes are verified from the cache and new entries are queued on the device, encrypted, and sent when the connection returns. A tablet offline for more than 15 minutes raises an alert to the estate manager.

15.4 Incidents and emergencies

- Incidents are categorised (intrusion, theft, fire, medical, dispute, damage, other) and timed, with photos.
- Serious incidents notify the estate manager and the agency supervisor at once by WhatsApp and SMS.
- Emergency broadcasts reach all residents in a block or the whole estate in one action.

SECTION 16

Process workflows

Workflows not covered in the preceding chapters. Steps marked **AUTO** run without user or staff action.

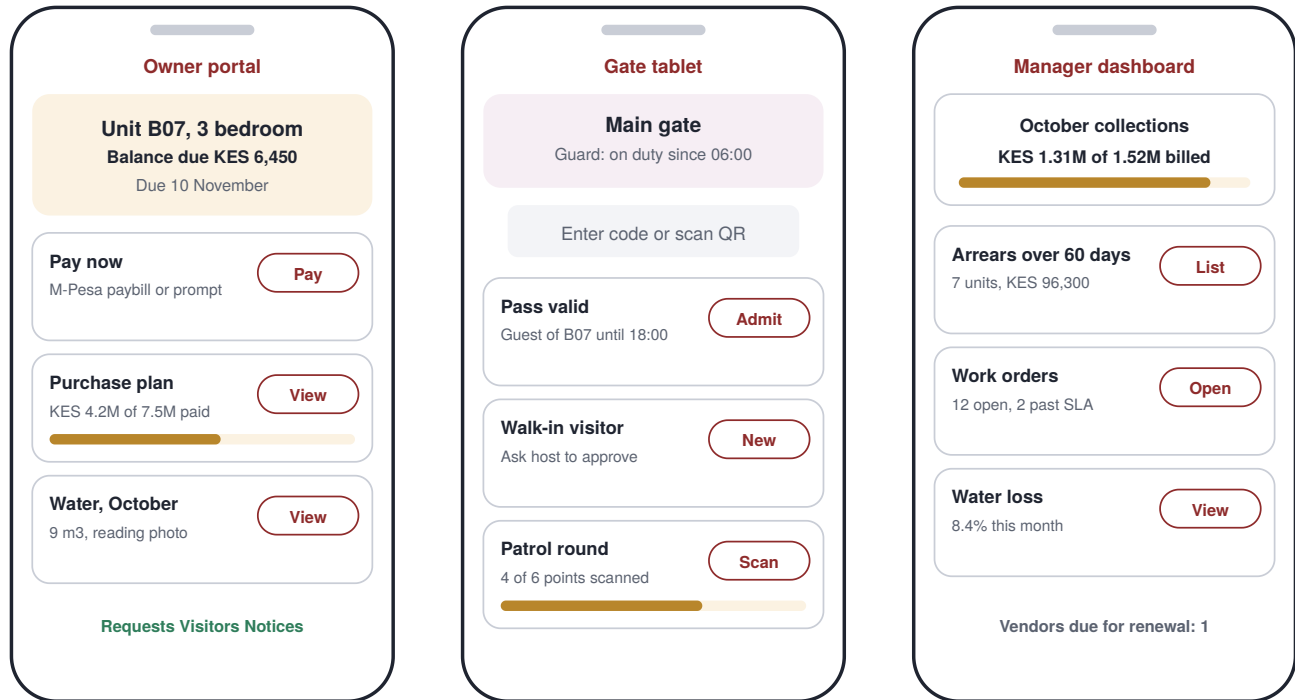


Figure 14. Key screens (low fidelity)

16.1 Launch tenant set-up: Shaba Village

Step	Action	Responsible
1	Create the Shaba Village tenant, assign the plan and create staff accounts with roles.	Codevertex
2	Connect the estate and sales paybills (Daraja C2B and prompt) and record both bank accounts. Map each to its ledger and cost centre.	Codevertex, Shaba finance
3	Configure charges, water tariff, penalties, allocation order, reminder days and approval limits.	Shaba finance
4	Import units, owners, occupants, sale contracts and opening balances; reconcile totals to Shaba's records before commit.	Codevertex, Shaba finance
5	Load vendors, contracts, licences, posts, checkpoints and preventive schedules.	Shaba management
6	Load estate staff into the ERP with contracts and pay details.	Shaba management
7	Run one billing cycle in parallel with the current method and compare unit by unit before going live.	Codevertex, Shaba finance
8	Live KES 1 tests on both paybills, and a test vendor payment, each reconciled.	Codevertex, Shaba finance

16.2 Owner onboarding

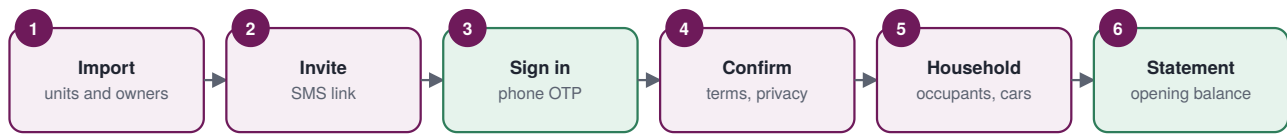


Figure 15. Owner onboarding

Step	Action	System behaviour
1	Owner records imported	Owner linked to unit with ownership date and opening balance
2	Invitation sent	SMS with portal link and paybill details AUTO
3	Owner signs in	One-time password by SMS, WhatsApp fallback; codes hashed, 5-minute expiry, rate limited
4	Owner accepts terms and privacy notice	Version and time recorded
5	Owner adds household members, tenants, vehicles and domestic staff	Gate access created for each
6	Owner views statement	Opening balance and current invoices from treasury

Exception: owners who do not use smartphones continue to receive bills and receipts by SMS and pay by paybill. Staff can create visitor passes on their behalf.

16.3 Unit resale or transfer

1. The selling owner requests a clearance certificate.
2. The platform checks that all estate balances are nil and issues the certificate with a reference and expiry. **AUTO**
3. On the transfer date, the old ownership closes, its account is finalised, and the new owner is invited. Meter readings are taken on the day so water is split correctly.
4. Service charge for the month is apportioned by days between the two owners.

16.4 Owner letting to a tenant

1. The owner adds the tenant as an occupant with start and end dates.
2. The owner chooses which charges the tenant receives (water, garbage, parking).
3. The tenant signs in by phone, creates visitor passes, raises requests and pays assigned bills with the unit account number.
4. On the end date, the tenant's access ends automatically. A final meter reading is requested. **AUTO**

16.5 Notices and alerts

1. The estate manager writes a notice, selects the audience (estate, block, owners, occupants) and channels.
2. Emergency alerts are sent immediately on all channels. Routine notices respect quiet hours, 21:00 to 07:00.
3. Delivery status per recipient is shown; failed SMS falls back to WhatsApp where available.

16.6 Annual budget and AGM

1. Finance prepares the next year's budget by cost line in treasury, with the prior year's actuals beside it.
2. The proposed service charge per unit type is calculated from the budget and unit entitlements.
3. The AGM pack (notice, budget, income and expenditure statement, sinking fund statement) is published in the document library and notified to owners.
4. Rates approved at the AGM are entered with their effective date and apply from the next billing run.

16.7 Handover to the management corporation

1. The estate fund ledger, unit accounts, sinking fund balance, vendor contracts, asset register and documents are exported.
2. The corporation is set up as its own tenant, or Shaba Village continues as managing agent under a new mandate.
3. Paybills and bank accounts are transferred or mandates changed, and opening balances are carried over and reconciled.

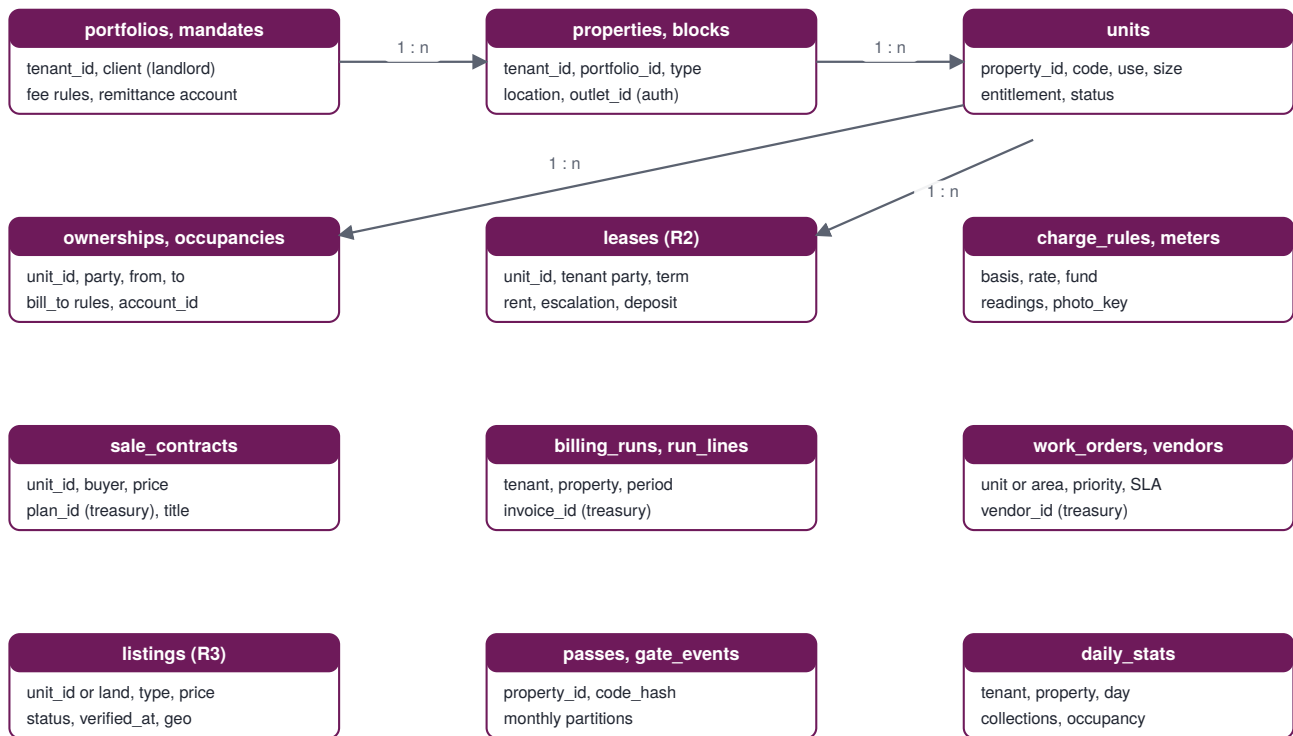
16.8 Data subject requests

1. An owner, occupant, visitor or agency worker submits a request for access, correction or deletion.
2. Identity is confirmed by one-time password and the statutory deadline is recorded.
3. Access requests produce a machine-readable export.
4. Deletion erases or anonymises personal data. Financial, sale and tax records required by law are retained for the statutory period, and the requester is told what is kept and why.

SECTION 17

Data design

property-api owns the multi-tenant property domain and references other services by ID: tenants, branches and users in auth-api; accounts, invoices, plans and vendors in treasury-api; employees and assets in erp-api; leads in marketflow-api.



Every table carries tenant_id; row-level security enforces it in the database.

Figure 16. property-api data model

17.1 Data owned elsewhere

Data	Owner	Held in property-api
Login identity, sessions, roles	auth-api	User ID
Invoices, payments, balances, statements, instalment plans, vendor bills, ledger	treasury-api	Account, invoice, plan and vendor IDs
Staff records, attendance, payroll, casual payments, assets	erp-api	Employee and asset IDs
Messages and delivery status	notifications-api	Message ID where audit requires

17.2 Scalability and performance measures

Measure	Purpose
Tenant and estate ID with query-specific composite indexes on every table, versioned Atlas migrations	Consistent query speed as data grows; controlled schema change
Keyspace pagination on all lists	Constant response time regardless of page depth
Gate events and patrol scans in monthly partitions, purged by partition at the end of retention	Fast writes and cheap deletion for the fastest-growing tables
Unique keys on (estate, period) for billing runs, (unit, period) for readings and run lines, and (event ID) for consumed events	No duplicates from retries or redelivered events
Daily aggregate table per estate, updated by events and rebuilt nightly	Dashboards and reports independent of raw data volume
Read replica for reports and exports; dashboard results cached for 60 seconds and invalidated by events	Reporting never slows the gate or payments
Bulk operations (billing, reminders, imports) in batches of 500 with progress tracking	Predictable memory and time at any estate size
Identity numbers and KRA PINs encrypted at field level; phone numbers hashed for matching	Data protection
Photos and documents in object storage with signed, short-lived links; keys only in the database	Small, fast database; private files

SECTION 18

Interfaces, events and notifications

18.1 property-api endpoints

Area	Endpoints	Access
Register	GET, POST /units; GET, PATCH /units/{id}; /units/{id}/ownerships, /occupancies, /vehicles; POST /imports	Staff; owners for own unit
Charges and billing	GET, PUT /charge-rules; POST /billing-runs; GET /billing-runs/{id}; POST /adjustments	Finance
Meters	GET /reading-rounds/{period}; POST /meters/{id}/readings; GET /water-balance	Caretaker, finance
Sales	GET, POST /sale-contracts; POST /sale-contracts/{id}/schedule, /restructure, /handover, /title-stages	Sales; buyer read-only
Owner portal	GET /me/units, /me/statement, /me/purchase; POST /me/passess, /me/requests, /me/queries	Owner, occupant
Works	GET, POST /work-orders; PATCH /work-orders/{id}; POST /work-orders/{id}/complete, /confirm, /reopen	Staff, vendor, resident
Vendors	/vendors/{id}/contracts, /documents, /visits, /personnel, /rosters, /invoices	Staff; vendor for own records
Gate	POST /gate/verify, /gate/events, /gate/sync; /patrols/scans; /incidents	Registered gate device
Communication	POST /notices	Staff
Documents	GET, POST /templates; POST /documents/generate; POST /documents/{id}/accept, /upload-signed; GET /documents/{id}/link	By role and relationship
Portfolios (R2)	GET, POST /portfolios; /portfolios/{id}/mandate, /statements, /remittances	Manager; landlord read
Leasing (R2)	/applications; /leases; POST /leases/{id}/activate, /review, /renew, /notice, /move-out; /inspections; /turnovers	Letting, manager; lessee own
Listings (R3)	POST /units/{id}/listing; /listings/{id}/publish, /close; public GET /market/search, /market/listings/{slug}; POST /market/enquiries, /viewings, /applications	Lister; public, rate limited
Tenant settings	/settings/charge-catalogue, /catalogues/{name}, /custom-fields, /approval-rules, /reminders	Tenant administrator
Reports and privacy	GET /reports/{name}; POST /privacy/export, /privacy/delete	By role

18.2 Treasury interfaces used

Function	Treasury capability
Owner and buyer accounts	Customer accounts and balances, one per unit and fund
Monthly and one-off invoices, credit notes	Invoice creation (service to service), recurring schedules, credit notes

Payments	Payment intents with M-Pesa prompt; paybill C2B confirmations matched by account number
Instalments	Instalment plans linked to the sale contract
Reminders	Dunning configuration and reminder events
Vendors	Vendor records, vendor bills, approvals, payouts, withholding tax
Budgets and statements	Budget lines, owner statements, income and expenditure, trial balance

18.3 Events

Event	Publisher	Consumer action
treasury.payment.succeeded	treasury	property-api refreshes the unit balance view, updates purchase progress and requests a receipt
treasury.invoice.created	treasury	property-api marks the run line as issued; notifications sends the bill
treasury.dunning.reminder_sent	treasury	notifications delivers the reminder; property-api records it on the unit timeline
treasury.payout.completed	treasury	property-api marks the vendor invoice as paid and notifies the vendor
property.billing_run.completed	property-api	Finance summary; dashboard refresh
property.work_order.created, assigned, completed	property-api	Notifications to resident, assignee and manager
property.visitor.arrived	property-api	Host notification
property.incident.reported	property-api	Alert to manager and agency supervisor
property.vendor.document_expiring	property-api	Alert to manager and vendor
property.lease.activated, renewed, ended (R2)	property-api	Billing schedule starts or stops; listing closes or reopens; notifications
property.listing.published, enquiry.created (R3)	property-api	Search projection refresh; lead created in marketflow; lister notified
property.document.executed	property-api	Key-date reminders scheduled; parties notified
property.unit.handed_over	property-api	Estate billing starts; welcome message to the owner

Consumers are idempotent on event ID. Events are held in the outbox during broker outages and delivered on recovery.

18.4 Notifications

Message	Recipient	Channel
One-time password	Owner, occupant, vendor	SMS; WhatsApp fallback
Monthly bill, reminders, receipts	Bill-to party	SMS or WhatsApp; email copy
Instalment due, received, missed	Buyer	SMS or WhatsApp; email copy
Visitor pass and arrival	Visitor; host	SMS or WhatsApp
Work order updates	Requester, assignee	WhatsApp or SMS
Notices and emergency alerts	Selected residents	SMS, WhatsApp, email, in-app
Licence expiry, missed visits, SLA breaches, incidents	Estate manager, vendor	WhatsApp, email
Approval requests, daily finance summary	Finance, directors	Email, WhatsApp

SECTION 19

Reporting and analytics

Report	Audience	Content
COLLECTIONS		
Collections summary	Management, finance	Billed, collected and collection rate by month, charge type and block
Arrears ageing	Finance	Balance per unit in 0 to 30, 31 to 60, 61 to 90 and over 90 day bands, with last payment date
Payment channel report	Finance	Paybill, prompt and bank receipts; unmatched and suspense items
Owner statement	Owner, finance	Invoices, payments and balance per fund
SALES		
Sales position	Directors, sales	Units by status and type; sold value; collected and outstanding
Instalment receivables	Directors, finance	Expected receipts by month for the next 24 months; overdue instalments; buyers in default
Purchase statement	Buyer, sales	Price, payments with receipts, balance, schedule
Handover and title tracker	Sales	Units awaiting handover, snag items open, title stage per unit
UTILITIES AND OPERATIONS		
Water consumption and balance	Management	Consumption by unit and block, estimated readings, unaccounted water percentage trend
Maintenance performance	Management	Work orders by category, time to respond and resolve, SLA breaches, cost by asset
Vendor scorecard	Management	Visits delivered and missed, post coverage, incidents, complaints, service credits
Security summary	Management	Visitors by day and hour, incidents by type, patrol completion
LETTINGS AND PORTFOLIOS (R2)		
Rent roll and occupancy	Manager, landlord	Units, lessees, rent, lease dates, vacancy and days vacant per property
Lease expiry and review schedule	Manager	Leases ending or due for review in the next 90 days
Deposits held	Finance, landlord	Deposits by lease, refunds due and settled
Landlord statement and remittance history	Landlord, finance	Collected, fees, expenses, tax, net remitted
Turnover performance	Manager	Days from move-out to ready, cost per turnover
MARKETPLACE (R3)		
Listing performance	Lister	Views, enquiries, viewings, applications and days to let or sell

PLATFORM

Tenant usage and health	Codevertex	Active tenants, units under management, collections volume, plan limits, failed jobs; no access to tenant detail
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FINANCE

Budget against actual	Management, owners (summary)	Each cost line, monthly and year to date
Income and expenditure, sinking fund statement	AGM	From the estate fund ledger
Trial balance, cash book, bank reconciliation	Finance, auditors	treasury financial reports for each fund

Reports run on aggregate tables and the read replica, never on the live transaction path. Each figure on a dashboard links to the records behind it, and totals reconcile to the ledger because invoices and payments are read from treasury rather than recalculated.

SECTION 20

Security

Threat	Controls
Account takeover	OTP rate limits by number and IP, hashed codes, short expiry, session revocation, MFA for staff, alerts on new device sign-in for finance roles
One resident seeing another's data	Every request checked against the caller's unit links on the server; IDs alone never grant access; tests for each role
Payment misallocation or fraud by staff	Adjustments and credit notes above a limit need a second approver; every allocation change logged; daily paybill reconciliation
Vendor payment redirection	Payout details verified and locked; changes need MFA and director approval; approvals bound to amount and payee
Duplicate invoices or payments	Unique run keys, receipt-number deduplication, idempotency keys on every write
Gate misuse	Tablets registered to the estate, guard PIN per shift, pass codes hashed and time-bound, one-time passes consumed on entry
Injection and cross-site scripting	Parameterised queries, output encoding, Content Security Policy, sanitised input and file type checks on uploads
Automated abuse	Rate limiting, STK prompt guard, Cloudflare WAF
Data exposure	TLS, encryption at rest, field encryption for identity numbers, least-privilege roles, signed short-lived file links, staff access logging

20.1 Production access

Control	Implementation
Who has access	A named list of Codevertex platform engineers, provided to tenants on request and updated within 5 working days of any change. No shared accounts.
Server access	Key-based SSH only, behind a default-deny firewall
Changes to production	Only through version-controlled GitOps deployment (ArgoCD); every change traceable to a reviewed commit
Secrets	Sealed secrets; separate database credentials per service; Daraja credentials encrypted in the database
Network isolation	Kubernetes network policies around payment, authentication and estate services
Tenant staff	Role-based access through single sign-on with MFA; approvals require a one-time password
Audit	Logins, approvals, settings and financial changes in tamper-evident audit logs, available to each tenant for its own data

20.2 Codevertex access to estate data

Codevertex staff do not access any tenant's personal or financial records in normal operation. Access occurs only to resolve a support request, investigate an incident or perform a restore, is limited to the records concerned, and is

logged. These obligations are set out in the data processing agreement, and every engineer with production access signs a confidentiality undertaking.

20.3 Security incident notification

Event	Timeline
Codevertex notifies the affected tenant of a confirmed personal data breach	Within 24 hours of confirmation
The tenant notifies the Data Protection Commissioner (section 43, Data Protection Act)	Within 72 hours of becoming aware; Codevertex supplies the facts
Affected people notified where there is a real risk of harm	Without undue delay, with the tenant's approval of the message
Written incident report: cause, impact, remediation	Within 5 working days

20.4 Penetration testing

An internal review against the OWASP Application Security Verification Standard is included in week 13. An independent penetration test is recommended before launch and annually, commissioned at the launch tenant's option and cost; Codevertex remediates all high and critical findings at no charge before launch.

20.5 Data processors

Processor	Purpose	Data received	Location
Codevertex Africa Limited	Software, hosting operation, support	All platform data	Kenya (operations); EU (servers)
Contabo GmbH	Server infrastructure	Platform data stored on its servers	Germany (EU)
Cloudflare, Inc.	Edge network, firewall, offsite backups	Traffic in transit; encrypted backup files	Global edge; EU and US
Safaricom PLC	M-Pesa collections and payouts	Payer number and name, amount, account number	Kenya
Africa's Talking	SMS delivery	Recipient number, message text	Kenya
Meta Platforms	WhatsApp messaging	Recipient number, message text	Global

SECTION 21

Hosting, backup and disaster recovery

Where tenant data is held, how it is backed up and how service is restored after a failure. Hosting is included in the monthly fee.

21.1 Hosting location

Component	Provider and location	Content
Application and database servers	Contabo Cloud VPS, European Union region (Germany), running Kubernetes (k3s)	property-ui, property-api, treasury, ERP, authentication, notifications; PostgreSQL databases
Offsite backups	Cloudflare R2 object storage, separate provider; encrypted	Nightly encrypted database backups
Edge network	Cloudflare (TLS, CDN, web application firewall)	Traffic in transit and cached public assets; no database content
Email	Codevertex mail server on the same platform	Outbound email; no third-party email processor
Money	Safaricom and each tenant's banks	Funds never reside on Codevertex infrastructure

Hosting outside Kenya is a cross-border transfer under sections 48 to 50 of the Data Protection Act. It is documented in the DPIA with its safeguards: provider security certification, encryption in transit and of backups, access restriction and a data processing agreement. Hosting in a Kenyan data centre is available on request and would be quoted as a change to the monthly fee.

21.2 Backup schedule

Backup	Frequency	Scope	Storage	Retention
Platform recovery backup	Daily, 05:00 EAT	All platform databases, verified after each run	Server disk and Cloudflare R2	Replaced only after the new copy is verified offsite
Per-tenant backup	Daily, 05:00 EAT	One tenant's property, treasury and ERP data, encrypted separately	Cloudflare R2, with optional mirror to the tenant's own storage (Google Drive, OneDrive, S3, SFTP)	7 daily and 4 weekly copies
Payment records	Continuous	Every M-Pesa collection and payout	Safaricom and the banks, independently of Codevertex	Provider retention

21.3 Recovery objectives

Objective	Target	Meaning for tenants
Recovery point objective (RPO)	24 hours	In the worst case, total loss of the server, data is restored from the last nightly backup. Payments made since then are recovered from M-Pesa statements and re-matched, so no payment is lost. Meter readings, work order updates and gate logs from that window may need to be re-entered or re-synchronised from the gate tablets.

Recovery time objective (RTO)	4 hours	Service is restored on a replacement server within 4 hours of a declared disaster.
Optional enhanced RPO	15 minutes	Continuous database log archiving to offsite storage, quoted on request.

21.4 Restore testing

A restore drill is performed quarterly: the latest offsite backup is restored into an isolated environment, record counts and ledger balances are compared with production, and a short report is shared with tenants on request. The first drill takes place before launch, in week 13.

21.5 Platform outage response

Scenario	Impact	Response
Service or pod failure	Seconds to minutes	Automatic restart; two replicas per service; health checks with alerting
Server failure or provider outage	Portal unavailable	Rebuild on a replacement server from version-controlled configuration, restore of the latest backup and re-matching of M-Pesa payments
During any outage	Owners cannot sign in	Paybill payments still reach each tenant's account; any confirmation missed during the outage is recovered from the M-Pesa statement and matched on recovery. Gate tablets continue offline with cached passes. Affected tenants are notified within 30 minutes of a confirmed incident and updated every 2 hours.

SECTION 22

Legal and regulatory compliance

The legal obligations that apply to the platform and its tenants, and the controls that address them. This is not legal advice. Items marked **Confirm** need an opinion from Shaba Village's advocate before launch.

22.1 Roles and responsibilities

Party	Role	Obligations
Each tenant (Shaba Village at launch)	Data controller for its properties, customers and staff; operator of its own estate, lettings or sales business	Agreements, by-laws, charges, collections, client money, vendor contracts, ODPC registration, tax, privacy notices, data requests
Codevertex as marketplace operator (R3)	Controller for public seeker and lister accounts and marketplace content	Lister verification, moderation, marketplace terms and privacy notice, takedown process
Codevertex Africa Limited	Software provider and data processor	Secure development and operation, processing on Shaba's instructions under a data processing agreement, breach notification
Service providers	Independent contractors; controllers of their own staff data	Licences, staff vetting and pay, statutory obligations, confidentiality of resident data they see
Safaricom, banks	Payment providers	Collection and payout of funds under their terms
Africa's Talking, Meta, email host	Sub-processors	Message delivery

22.2 Applicable law

Law	Obligation	Control
Constitution of Kenya, Article 31; Data Protection Act, 2019; General Regulations, 2021	Lawful, minimal and transparent processing; data subject rights; DPIA for high-risk processing; 72-hour breach notification; transfer safeguards; consent for marketing; parental consent for children's data	Privacy notices, data minimisation, consent records, DPIA, data request workflow, breach procedure, documented hosting, transactional messages only by default
Data Protection (Registration of Data Controllers and Processors) Regulations, 2021	Property managers, landlords and real estate businesses must register with the ODPC	Shaba Village registers before launch; Codevertex is registered as processor
ODPC draft Guidance Note for the Private Security Sector, 2025	Proportionate visitor data, transparency at gates, CCTV and access logs subject to the Act, retention limits	Name, phone, host and plate only; no ID copies; gate privacy notice; 90-day gate log retention; access by role
Sectional Properties Act, 2020	Management corporation for common property; service charge by unit entitlement; sinking fund; accounts and records; by-laws; AGMs	Entitlement-based charges, separate sinking fund ledger, estate fund books ready for handover, AGM pack, document library

Land Registration Act, 2012; Land Act, 2012; Law of Contract Act	Contracts for disposition of land in writing and signed; registration of transfers and charges	Signed agreements stored; conveyancing by advocates; title stages tracked, not performed
Consumer Protection Act, 2012	Fair terms; clear disclosure of charges and deposits; no misleading representations	Charge schedule and penalties published to owners; bills itemised with evidence; query window on every bill
Landlord and tenant law and case law of the Business Premises Rent Tribunal and courts	No self-help disconnection of water or electricity or interference with possession without due process	No automated disconnection or lock-out; arrears handled by reminders, plans and legal process
Water Act, 2016; WASREB; WRA	Abstraction permits for boreholes; water supply in a licensed provider's area; water quality	Permit and water test documents tracked with expiry; bulk and abstraction readings recorded. Confirm the basis on which Shaba supplies and bills water within MAVWASCO's area, and whether a bulk supply agreement is needed.
Energy Act, 2019; EPRA	Only licensed persons sell electricity	Units on Kenya Power prepaid meters; the estate does not resell power; common area power is an estate expense
Sustainable Waste Management Act, 2022	Segregation at source; collection only by NEMA-licensed providers	Collector licence held on file with expiry alert; segregated collection in the SLA
Private Security Regulation Act, 2016 and Regulations, 2019	Security services only by PSRA-licensed firms and registered officers	PSRA licence verified and tracked; only registered agency guards rostered
Estate Agents Act	Persons acting as agents for property sales or letting for others must be registered with the Estate Agents Registration Board, with an annual practising certificate	Property manager tenants and agent listers give their EARB number at onboarding, checked and re-checked yearly. Confirm for Shaba whether any staff act as agents for third parties.
Landlord and tenant law: Rent Restriction Act, Landlord and Tenant (Shops, Hotels and Catering Establishments) Act, Distress for Rent Act; pending Landlord and Tenant Bill	Controlled tenancies, notice requirements for rent changes and termination, tribunal jurisdiction, no self-help eviction or disconnection	Notice periods taken from the lease and configurable minimums; notices generated and logged; no lock-out or disconnection features; deposit refunds with itemised deductions
Land Registration Act, 2012; Stamp Duty Act	Leases over two years must be registered; leases and sale agreements attract stamp duty	Registration and stamp duty flagged on generation; references recorded
Income Tax Act: monthly rental income tax and withholding on rent	Resident landlords pay 7.5% on gross residential rent; appointed agents withhold 7.5%; non-resident landlords suffer 30%; commercial rent withholding by appointed lessees	MRI and withholding computed on landlord statements; PRNs generated through treasury's KRA integration; certificates stored
Competition Act, 2010; Consumer Protection Act (marketplace)	No false or misleading advertising of property, price or availability	Verified listers, listings closed when let or sold, price and photo moderation, takedown on report
Proceeds of Crime and Anti-Money Laundering Act, 2009, as amended	Real estate businesses carry customer due diligence, record keeping and reporting duties	Buyer KYC on each sale contract, source of funds field, records kept 7 years, cash payments recorded. Confirm Shaba's reporting obligations and thresholds.

Employment Act, 2007; Occupational Safety and Health Act, 2007; Work Injury Benefits Act, 2007	Contracts, pay records and statutory deductions for estate staff; safety of contractors on site	ERP payroll with statutory deductions; contractor insurance certificates tracked
Income Tax Act; VAT Act, 2013; Tax Procedures Act, 2015; eTIMS	Withholding tax on qualifying payments; eTIMS invoices for deductible expenses since 2026; records for at least 5 years	WHT computed on vendor payments; eTIMS number required on vendor invoices; 7-year retention. Confirm VAT treatment of service charge; residential letting and sale are exempt, and a July 2026 VAT Tribunal decision held owners' service charge contributions not taxable.
Kenya Information and Communications Act; Consumer Protection Regulations, 2010	No unsolicited marketing messages; opt-out	Estate messages are service messages; any marketing needs opt-in with STOP
Computer Misuse and Cybercrimes Act, 2018	Protection of systems and data	Security controls, audit logging, incident response
Children Act, 2022; Data Protection Act, section 33	Children's data only with parental consent and in their best interests	Household members under 18 added only by the owner, name only, for gate access
Persons with Disabilities Act; Constitution, Article 54	Accessible services	WCAG 2.2 AA
EU and UK GDPR (owners in the diaspora)	Equivalent privacy rights where offering services to people in the EU or UK	Kenyan controls applied to all owners; prior consent for optional cookies; documented transfers

22.3 Compliance checklist

#	Requirement	Implementation	Owner
1	Privacy notice	Controller, processors, purposes, lawful bases, retention, rights, transfers, contacts. Linked from every page and message; summary at each gate.	Codevertex draft; advocate approval
2	Owner portal terms	Use of the portal, payment references, bill-to assignment, queries, Kenyan law. Versioned acceptance.	Codevertex draft; advocate approval
3	Charge schedule and penalty policy	Published from by-laws and agreements; matches platform settings.	Shaba Village
4	Gate and CCTV notice	Purpose, data collected, retention, contact.	Codevertex draft
5	Cookie consent	Strictly necessary cookies only at launch; equal accept and reject.	Codevertex
6	Data minimisation	No ID copies at the gate; identity documents only for sale contracts, encrypted.	Codevertex
7	Role-based access	Sales, finance, caretaker, guard and vendor roles see only what they need.	Codevertex
8	Vendor confidentiality	Clause in every provider contract covering resident data seen on site.	Shaba Village
9	No dark patterns	No pre-ticked consents; leaving a pass or request is as simple as creating one.	Codevertex

10	Itemised bills with evidence	Every metered charge shows readings and photo.	Codevertex
11	Accessibility	Contrast 4.5:1, keyboard operation, alternative text, automated checks.	Codevertex
12	Business details	Registered name, address and contacts on every bill, receipt and statement.	Codevertex
13	Children's data	Owner-only entry of minors, name only.	Codevertex
14	Retention and deletion	Schedule below enforced by automated purge; data requests tracked to deadline.	Codevertex
15	Registers	Record of processing, processor list, licence register for fonts and images.	Codevertex

22.4 Record retention

Record	Retention	Basis
Invoices, payments, ledger, vendor bills, payroll	7 years	Tax Procedures Act; AML record keeping
Sale contracts, buyer KYC, title records	7 years after completion of transfer	Contract and AML record keeping
Leases, mandates, inspection reports, deposit settlements	7 years after the lease or mandate ends	Limitation of actions; tax records
Unsuccessful rental applications and their documents	6 months after decision	Data minimisation
Marketplace enquiries and viewing records	12 months	Service delivery and fraud investigation
Closed listings	24 months, then anonymised	Duplicate and fraud detection
Owner and occupant accounts	While linked to a unit; then 24 months, then anonymised	Data minimisation
Meter readings and photos	Readings 7 years; photos 24 months	Billing evidence
Gate events and visitor details	90 days	Security; ODPC guidance
Incident reports	3 years, or longer if under investigation or claim	Security and insurance
Agency personnel records	12 months after deployment ends	Access control
Work orders	7 years	Asset history and cost evidence
Consent and terms acceptance	Account life plus 5 years	Evidence of consent
One-time passwords	5 minutes, hashed	Security
Security and audit logs	12 months	Incident investigation

22.5 Legal deliverables

- Privacy notice, gate notice, cookie policy, owner portal terms, vendor portal terms; lessee, landlord and marketplace terms for later releases
- Starter document templates for each tenant type, for review by each tenant's advocate
- Data processing agreement and sub-processor list
- Data protection impact assessment and record of processing activities
- Breach response and data request procedures
- Vendor confidentiality clause, accessibility statement

SECTION 23

Project delivery

Delivery follows Agile Scrum with weekly demonstrations on the staging environment.

23.1 Methodology

Practice	Approach
Iterations	One-week Sprint 0, then two-week sprints, a hardening week and a launch week
Demonstrations	Weekly 30-minute review with the product owner; decisions logged
Backlog	User stories traced to FR and NFR identifiers, with acceptance criteria
Definition of done	Code reviewed, tests and accessibility checks passed, documented, deployed to staging, accepted
Data migration	Import template issued in week 1; trial imports from week 3; balances reconciled with Shaba before go-live
Parallel run	One billing cycle produced by both the old method and the platform, compared unit by unit
Release management	CI on GitHub Actions; GitOps deployment through ArgoCD
Change control	Equal-size scope substitutions within the fixed price; other changes estimated and approved in writing

23.2 Project team

Role	Allocation
Project lead and business analyst	Part time, weeks 1 to 14
Full-stack engineers (Go, Next.js)	Two, weeks 1 to 14
Product designer	Part time, weeks 1 to 5; reviews thereafter
QA engineer	Part time, weeks 3 to 14
Platform and DevOps engineer	As required

23.3 Implementation schedule

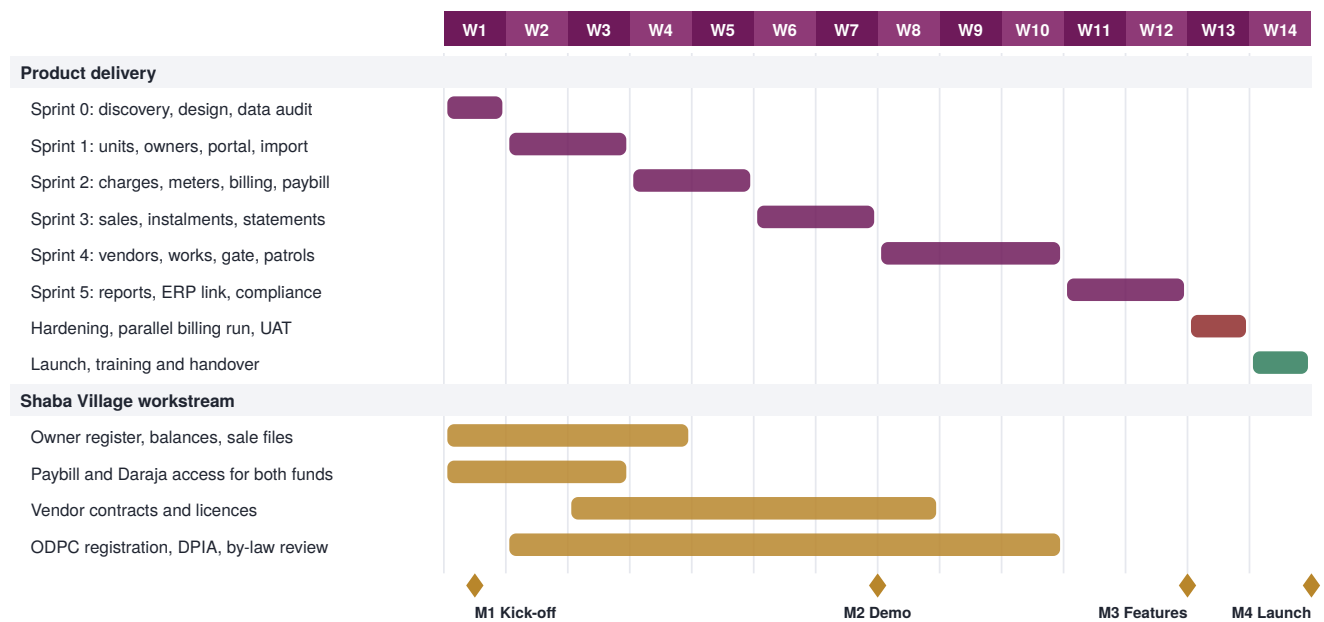


Figure 17. Implementation schedule

Week	Phase	Deliverables
1	Sprint 0	Kick-off, by-law and agreement review, data audit and import template, UX flows, visual design, architecture approval, environments
2 to 3	Sprint 1	Units, owners, occupants, phone sign-in, owner portal shell, imports, staff roles
4 to 5	Sprint 2	Charge rules, meter readings, billing runs, paybill matching and allocation, receipts, statements, reminders
6 to 7	Sprint 3	Price lists, reservations, sale contracts, instalment schedules, purchase statements, handover. Milestone 2: end-to-end demonstration with live paybill payments
8 to 10	Sprint 4	Vendors, contracts, schedules and evidence, vendor portal and invoices, work orders, gate tablet, passes, patrols, incidents
11 to 12	Sprint 5	Reports and dashboards, budgets, ERP staff and casual payments, notices and documents, privacy tools, accessibility. Milestone 3: feature complete
13	Hardening	Security review, load test, restore drill, parallel billing run, UAT, legal pack approval
14	Launch	Production release, owner invitations, training, handover. Milestone 4: launch acceptance

Based on contract signature by 23 October 2026, work begins on 26 October 2026. Weeks 1 to 8 run to 18 December. The team pauses from 21 December to 3 January, and work resumes on 4 January 2027 with launch in the week of 8 February 2027.

23.4 Later releases

Release	Indicative timing	Pilot
R2 Rental and portfolio management, self-service tenant sign-up	Development March to May 2027; general availability June 2027	One property management firm and one self-managing landlord
R3 Public marketplace	Development June to August 2027; launch September 2027, seeded with vacancies from managed properties	Shaba Village resales and R2 tenants' vacancies
R4 Extensions	From Q4 2027, by demand	As agreed

SECTION 24

Testing and acceptance

Test	Objective	Exit criteria
Automated tests	Charge calculation, proration, allocation, instalments, SLA credits	All tests pass on every merge
End-to-end tests	Owner, buyer, caretaker, guard, vendor and finance journeys	Pass on Android Chrome, iOS Safari and desktop
Migration reconciliation	Imported balances and contracts	Totals per unit and fund match Shaba's signed-off figures
Parallel billing run	One month billed both ways	Every unit's bill matches, or the difference is explained and accepted
Live payment tests	Both paybills, prompt, vendor payout	KES 1 transactions matched, receipted and reconciled
Offline gate test	Network removed for 2 hours	Passes verified, entries queued and synchronised without loss
Load test	Peak use	Billing run for 1,000 units within NFR-06; 50 requests per second within NFR-04
Security review	OWASP risks, access between units, payment adjustments	No open high or critical findings
Tenant isolation tests	Two test tenants with overlapping data; every route, export, file link, event and report checked across tenants	Zero cross-tenant results, run on every release
Configuration tests	Each use case preset and module switched on and off	No screen, route, job or message of a disabled module reachable; data kept on switch-off
Accessibility	WCAG 2.2 AA	Automated and manual checks passed
User acceptance	Business acceptance by Shaba Village staff and pilot owners	All M requirements accepted; no open severity 1 or 2 defects

SECTION 25

Commercial terms

25.1 Fixed price: Release 1 and Shaba Village launch

Work package	KES
Discovery, by-law and data review, UX and visual design, architecture	15,000
Multi-tenant core: tenancy and isolation, modules and use cases, charge and other catalogues, layered configuration, plans and limits	30,000
Property service: register, staff per property, billing runs, meters, sales and instalments, works, vendors, gate and visitors	60,000
Web application: owner portal, management console, gate tablet, vendor portal, configuration-driven navigation and forms	55,000
Finance integration: invoices, paybill matching, allocation, instalment plans, statements, reminders, vendor bills, payouts, fund ledgers	35,000
Documents and templates, ERP set-up (staff, payroll, casual payments), message templates	15,000
Shaba data migration, compliance features, legal pack drafts	20,000
Testing, security and isolation review, parallel run, UAT, training, launch	20,000
Total (inclusive of taxes)	250,000

The ERP implementation fee normally charged on a new ERP subscription is included. Releases 2 to 4 are developed by Codevertex as part of the platform roadmap and are not charged to Shaba Village.

25.2 Payment schedule

Milestone	Due	Share	KES
M1 Contract signature	Week 0	30%	75,000
M2 End-to-end demonstration	End of week 7	30%	75,000
M3 Feature complete	End of week 12	30%	75,000
M4 Launch acceptance	End of week 14	10%	25,000
Total		100%	250,000

25.3 Monthly service tiers

The monthly fee follows the Codevertex ERP subscription tiers and applies to every tenant type on the platform, measured by units under management. Each tier includes the Lintel platform licence, the ERP and treasury subscription at that tier, hosting, backups, monitoring, security updates, and support and maintenance. Modules released later (leasing, landlord portfolios, marketplace publishing) become available to subscribed tenants within their tier at no extra charge.

Tier	Units	Estate staff in ERP	Staff users	KES per month
Starter	Up to 150	25	10	10,000
Growth (recommended for Shaba)	Up to 400	75	30	20,000
Professional	Up to 1,000, several properties or phases	200	100	35,000
Enterprise	Unlimited	Unlimited	Unlimited	By quote

Owners, occupants, guards and vendor supervisors are not counted as staff users and are unlimited in every tier.

Capability	Starter	Growth	Professional
Register, billing, collections, sales and instalments, owner portal, gate, vendors, work orders, notices	Yes	Yes	Yes
Payroll, casual payments, leave, attendance	Yes	Yes	Yes
Multi-level approvals for vendor bills, credit notes and restructures	Single level	Yes	Yes
Service charge budget against actual, asset register, BI reports	No	Yes	Yes
API access, custom workflows, extended audit trail, priority support	No	No	Yes

Recommendation for Shaba Village: Growth at KES 20,000 per month. The budget against actual report (FR-65), multi-level approvals (FR-29, FR-57) and the asset register used by preventive maintenance (FR-58) are Growth features, and they carry most of the transparency owners expect from estate management.

Term	Position
Billing	Monthly in advance from the launch month. Annual prepayment earns one month free.
Changing tier	Upgrades apply at once; downgrades from the next month if usage fits the lower tier.
Usage limits	No per-transaction or per-payment charge from Codevertex.
Price changes	Fixed for 12 months from launch; thereafter at most 10% a year, with 60 days' written notice.

25.4 Client-borne costs

Item	Scope	Rate (KES)
SMS	Bills, receipts, reminders, passes, one-time passwords; prepaid, minimum top-up KES 500	1.00 per SMS
WhatsApp	Bills, receipts, work order updates, alerts; monthly plan	500 to 2,000 per month
M-Pesa charges	Safaricom paybill and payout tariffs on Shaba's own paybills	Safaricom tariff
Domain	Registration and renewal of Shaba Village's portal domain	Registrar rate
Gate tablets and patrol tags	One Android tablet per gate with data; weatherproof QR tags for checkpoints	Supplier rate

25.5 Messaging rates

Plan or item	Allowance	KES
SMS	Per SMS of up to 160 characters	1.00
WhatsApp Basic	100 messages per month	500 per month
WhatsApp Standard	500 messages per month	1,000 per month
WhatsApp Pro	Unlimited messages	2,000 per month

25.6 Operating cost examples

Monthly item (KES), Growth tier	120 units	250 units	400 units
Monthly service fee	20,000	20,000	20,000
SMS (5 per unit: bill, reminder, receipt, passes)	600	1,250	2,000
WhatsApp Pro	2,000	2,000	2,000
Total	22,600	23,250	24,000
Cost per unit per month	188	93	60

M-Pesa charges depend on the tariff on Shaba's paybills and are excluded. The cost can be recovered within the management line of the service charge budget.

25.7 Optional services

Service	Scope	Billing
Managed domain	Registration, renewal, DNS and certificates	Annually, at the registrar's rate
Smart water meters	Integration of remote-read meters through the Codevertex IoT service	Quoted on meter choice
Letting service for Shaba owners	Shaba managing lettings for owners who rent out their units, using the Release 2 leasing and landlord modules	Included in tier when released; Shaba sets its own management fee
Kenyan data centre hosting	Hosting within Kenya	Quoted as a change to the monthly fee

SECTION 26

Ownership, licensing and exit

Shaba Village owns its data, documents and money. Codevertex owns the platform, which Shaba Village uses under licence.

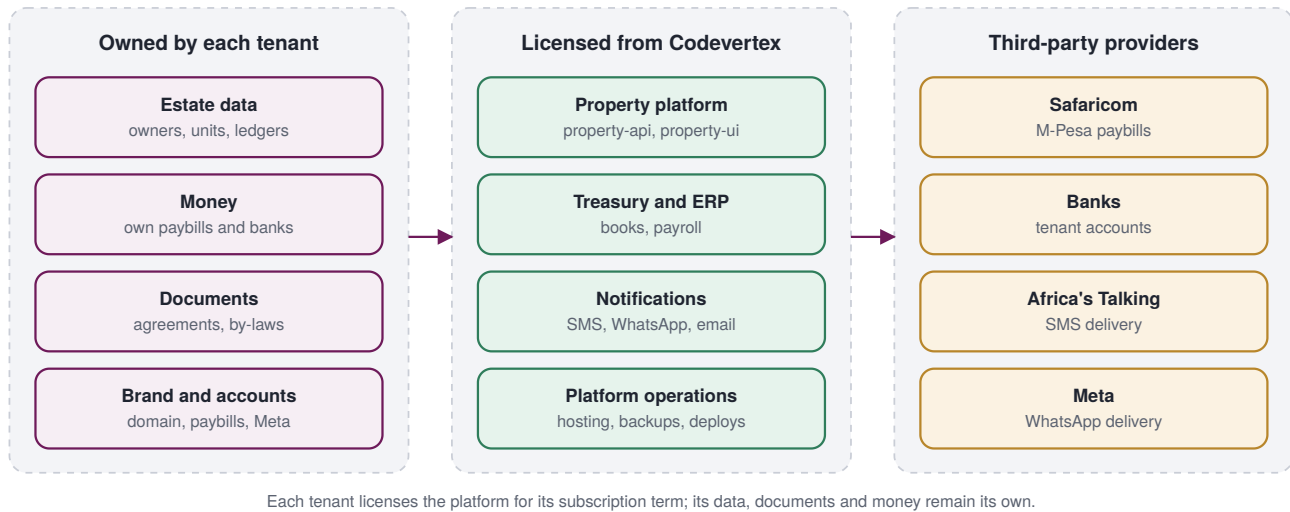


Figure 18. Ownership boundaries

26.1 Ownership

Asset	Owner	Terms
Estate data: units, owners, occupants, readings, invoices, payments, contracts, work orders, gate records, ledgers	Shaba Village	Owned from creation; Codevertex processes it only on Shaba Village's instructions
Shaba Village configuration, templates, documents, designs and content	Shaba Village	Assigned to Shaba Village on payment of the build price
Money	Shaba Village and, for the estate fund, the owners	Held in Shaba's paybills and bank accounts, never by Codevertex
Brand, domain and accounts with Safaricom, banks and Meta	Shaba Village	Registered in Shaba Village's name
Codevertex platform: property service, marketplace, authentication, treasury, ERP, notifications, subscriptions, shared libraries, deployment tooling	Codevertex	Proprietary; licensed as below

26.2 Platform licence

Shaba Village receives a non-exclusive, non-transferable licence to use the platform for the duration of the service agreement, included in the monthly fee. Shaba Village's own developers, or another vendor it appoints, may use the documented APIs with keys issued to Shaba Village. The licence passes to the management corporation, at Shaba's direction, on handover of the estate.

26.3 Exit and migration

Step	Commitment
Notice	At least 90 days' written notice before Codevertex discontinues any service Shaba Village relies on.
Data export	Complete export in open formats: database dump of Shaba Village's records, ledgers and statements (CSV and Excel), unit and owner records (JSON), photos and documents.
Money	Unaffected. Funds are in Shaba's own paybills and bank accounts.
Transition assistance	Up to 20 hours of engineering support, included; further support at the standard rate.
Source code escrow (optional)	Source code of the services Shaba Village relies on deposited with an independent escrow agent, released if Codevertex ceases to trade. Agent fees borne by Shaba Village.

26.4 Handover on termination

- Final data export and the latest backups within 10 working days
- Operations documentation, the legal pack and registers
- Transfer of domain and DNS administration, where managed by Codevertex
- Written confirmation of deletion of Shaba Village data from Codevertex systems after export, subject to statutory retention

SECTION 27

Contract terms

Principal terms of the master services agreement between Shaba Village and Codevertex. Codevertex's standard terms apply where not varied below.

Term	Position
Initial term	12 months from launch, renewing monthly thereafter
Termination for convenience	Either party, after the initial term, with 60 days' written notice
Termination for cause	Either party, for a material breach not remedied within 30 days of written notice, or on insolvency
Fee after termination	Payable to the end of the notice period only. Export and handover at no charge.
Service level	99.5% monthly availability for the owner portal, gate service and payment matching, excluding maintenance notified 48 hours ahead
Service credits	Below 99.5%: 10% of the monthly fee. Below 99.0%: 25%. Below 95.0%: 50%.
Prolonged downtime	An outage over 72 continuous hours, or service credits in 2 consecutive months, entitles Shaba Village to terminate without notice and receive full handover.
Delay caused by Codevertex	2% of the build price credited for each full week of delay to launch caused solely by Codevertex, up to 10% (KES 25,000). The final milestone is not due until launch acceptance.
Delay caused by Shaba Village	Where data, decisions, documents, paybill access, feedback or test participation are later than agreed (decisions within 2 working days, documents within 3), the schedule extends day for day with no delay credits. Milestones fall due on their original dates for work completed. Team standby beyond a cumulative 5 working days is charged at KES 20,000 per week, pro rata.
Force majeure and third parties	Events outside both parties' control, including delays by Safaricom or banks after complete documents were submitted on time, extend the schedule day for day with no credits or charges. Beyond 60 days, either party may terminate, and Shaba Village pays for work completed.
Liability cap	Each party's total liability is limited to the fees paid in the 12 months before the claim. Indirect and consequential losses are excluded.
Exclusions from the cap	Breach of confidentiality, the intellectual property indemnity and wilful misconduct
Codevertex indemnity	Against third-party claims that the software infringes intellectual property, and regulatory fines arising from Codevertex's breach of the data processing agreement, within the cap
Shaba Village indemnity	Against claims arising from Shaba Village's charges, by-laws, sale agreements, content and handling of funds
Governing law	Laws of Kenya. Negotiation for 30 days, then mediation, then the courts of Kenya.

SECTION 28

Risks and mitigations

Risk	Impact	Mitigation	Owner
Incomplete or disputed opening balances	Owner complaints at launch	Data audit in week 1, trial imports, reconciliation sign-off, statement review by owners in the first month	Shaba Village, Codevertex
Paybill or Daraja access delayed	Payments not matched automatically	Requested in week 1; manual statement import as fallback	Shaba Village
Water bill disputes	Collection delays	Photo per reading, anomaly checks, query window, meter replacement records	Shaba Village operations
Owners not adopting the portal	Reliance on office queries	SMS bills and paybill work without the portal; launch communication; office assisted sign-up	Shaba Village
Provider resistance to SLA checks	Incomplete evidence	SLAs written into renewed contracts; simple check-in on any phone; supervisor training	Shaba Village
Gate network failure	Slow entry	Offline mode with cached passes; device alerts	Codevertex
Legal position on water supply or agents	Change to billing or roles	Advocate review from week 2; configurable charges	Shaba Village
Cross-tenant data exposure	Severe legal and trust damage	Row-level security, server-side scoping, automated isolation tests, security review each release	Codevertex
Configuration grows into custom code	Harder upgrades for all tenants	Every tenant need met by catalogue, setting or module; product review before any new option	Codevertex
Fraudulent marketplace listings (R3)	Losses to seekers, reputation	Lister verification, managed-vacancy sourcing, moderation, masked contacts, safe payment rule	Codevertex
Scope change	Schedule slippage	Change control; agreed exclusions	Both
Holiday period	Slower decisions	Planned pause; decisions scheduled before 18 December	Both

SECTION 29

Post-launch support

Service	Terms
Hypercare	30 days after launch, covering the first full billing cycle; same-day response to billing and payment incidents
Warranty	Defects against this specification corrected at no cost for 90 days
Ongoing support	Support and maintenance included in the monthly fee: defect correction, configuration help, platform updates
Response targets	Critical (payments, gate, portal down): 1 hour. High: 4 working hours. Normal: next working day.
Monitoring	Alerts on failed billing runs, unmatched payments, offline gate tablets, error rates and latency
Training	Management and finance (two sessions), caretakers, guards and vendor supervisors (one session each), with short video guides
Handover	Administration guide, finance guide, gate guide, runbooks, legal pack, registers
Enhancements	Smart meters, rent collection for owners, owner voting and native applications quoted on request

SECTION 30

Acceptance

Signature below confirms acceptance of the scope, schedule and price in this document. Changes are subject to the change control process in [Project delivery](#).

Shaba Village	Codevertex Africa Limited
Name	
Title	
Signature	
Date	